



Voronoi-Based K Nearest Neighbor Search for Spatial Network Databases

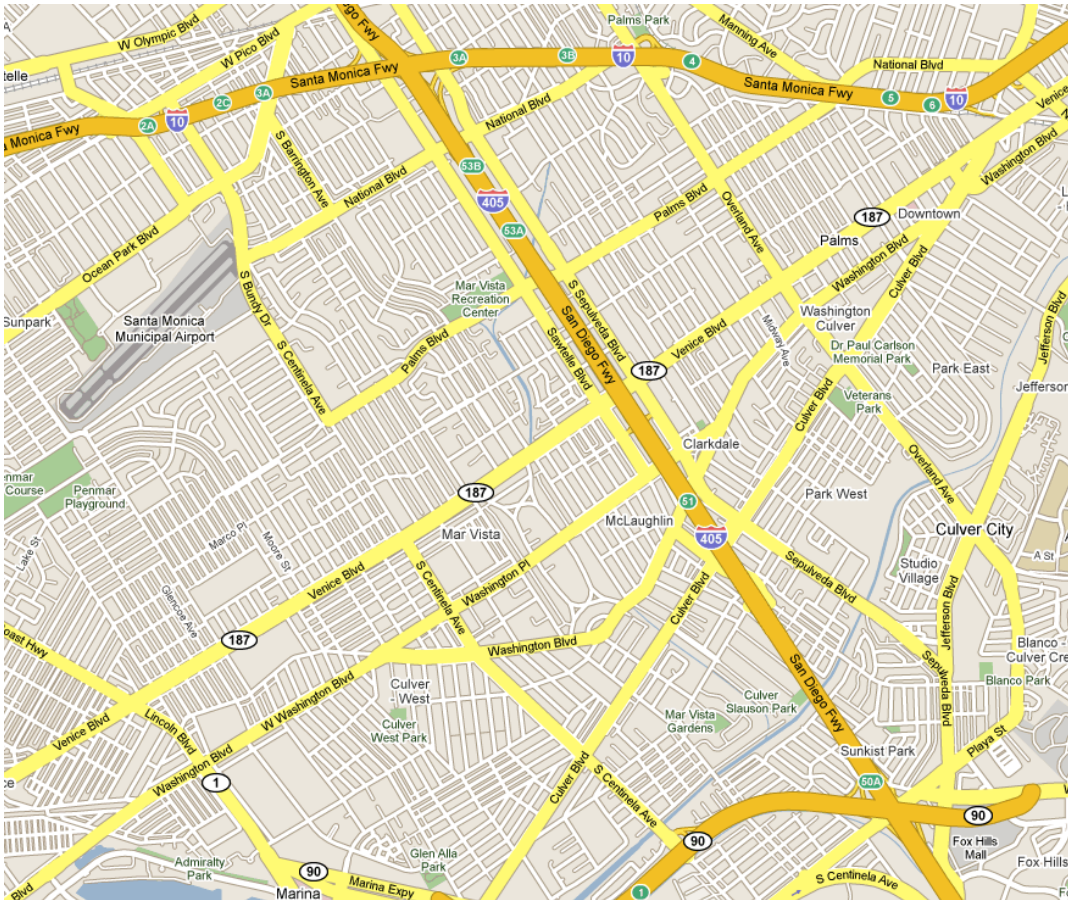
Instructor: Cyrus Shahabi



Agenda

- Problem Definition
- Related Work
- Voronoi Diagrams
- Voronoi Based Network Nearest Neighbor (VN³)
- Experiments
- Discussion

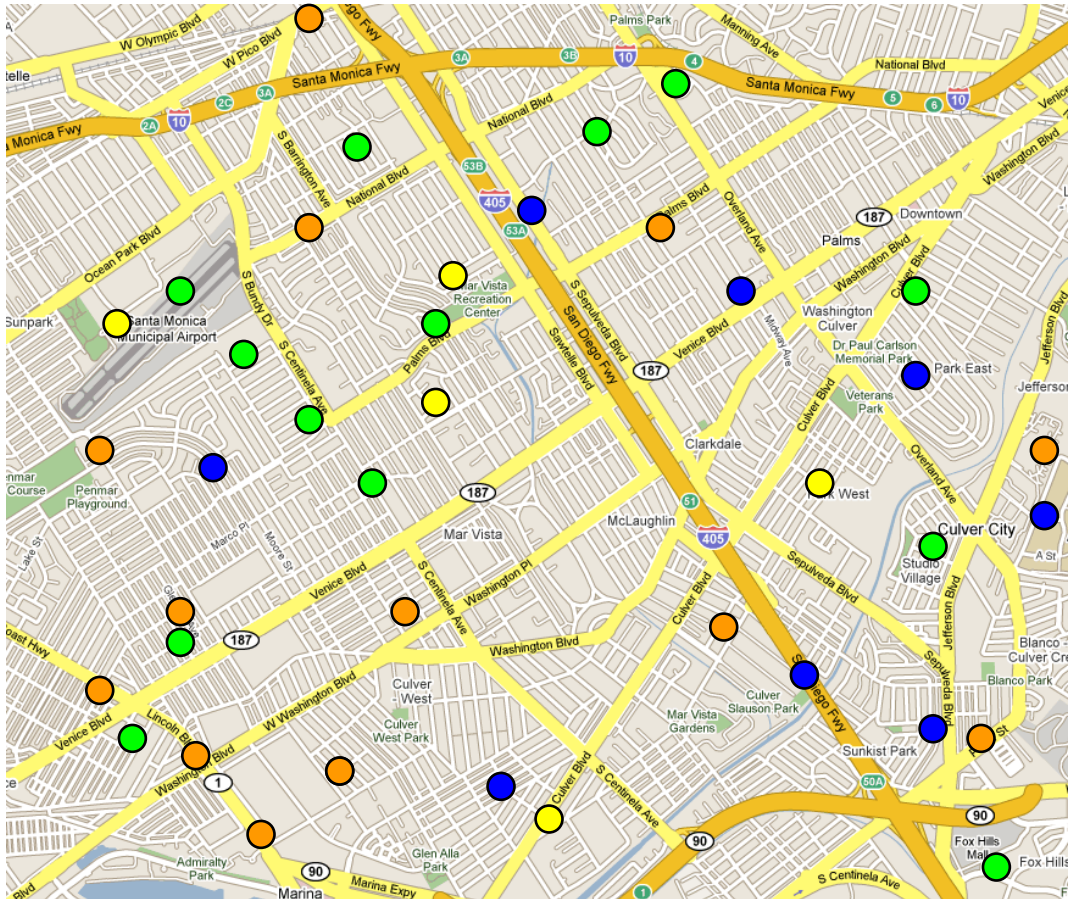
Problem Definition



kNN Problem:

Given a set of objects P and query point q , find the k objects in P that are closest to q

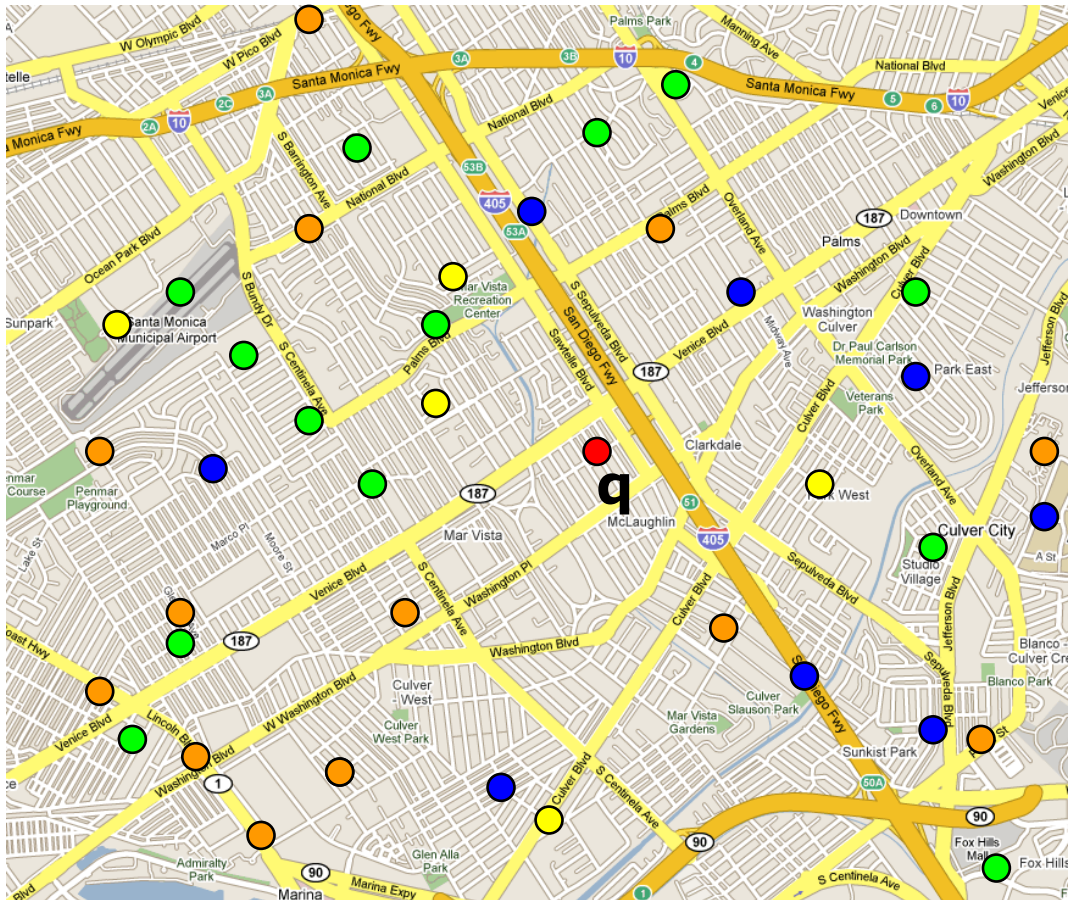
Problem Definition



kNN Problem:

Given a set of objects P and query point q , find the k objects in P that are closest to q

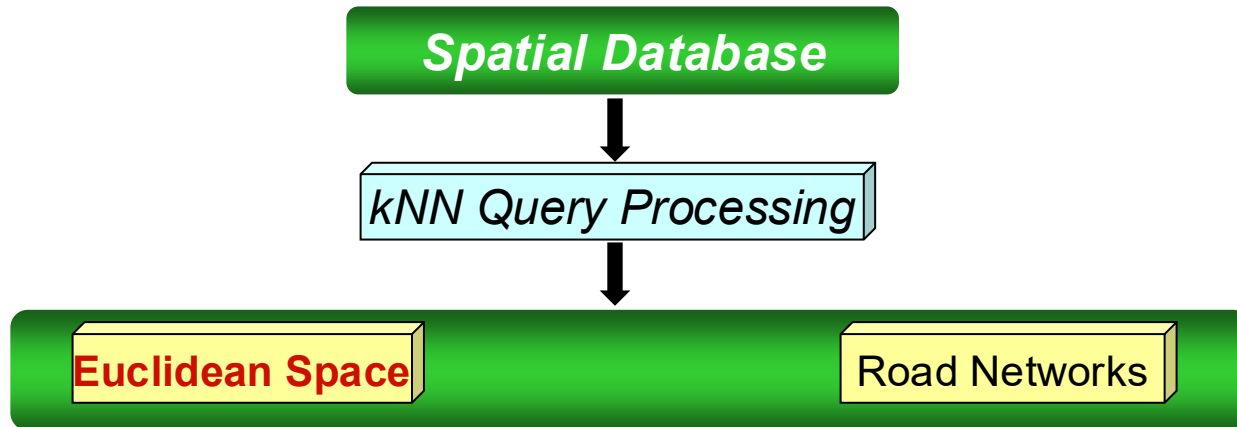
Problem Definition



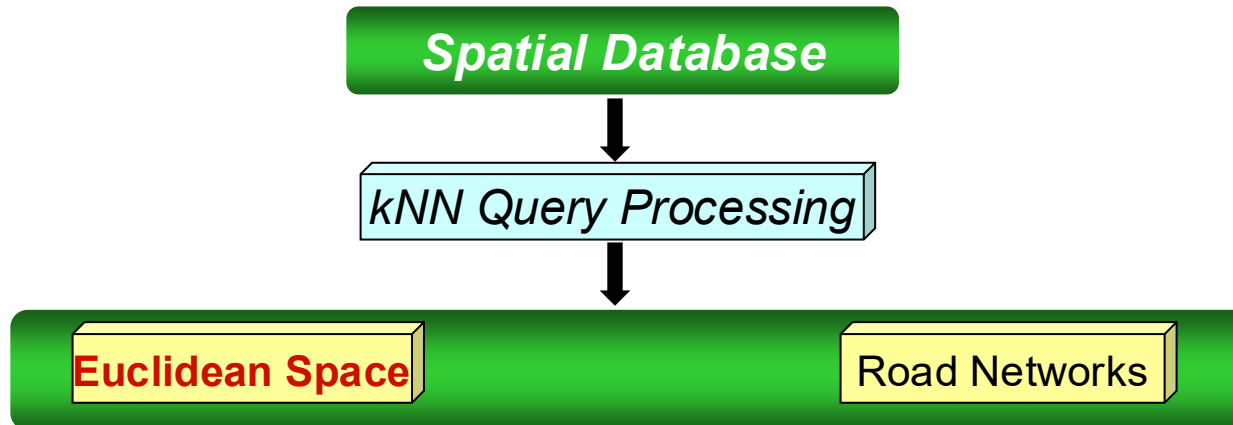
kNN Problem:

Given a set of objects P and query point q , find the k objects in P that are closest to q

Related Work



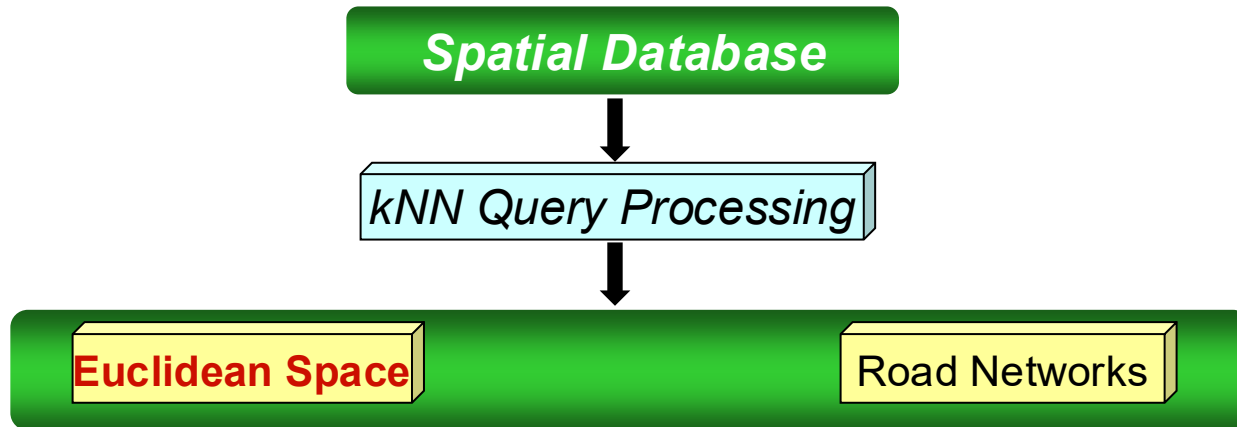
Related Work



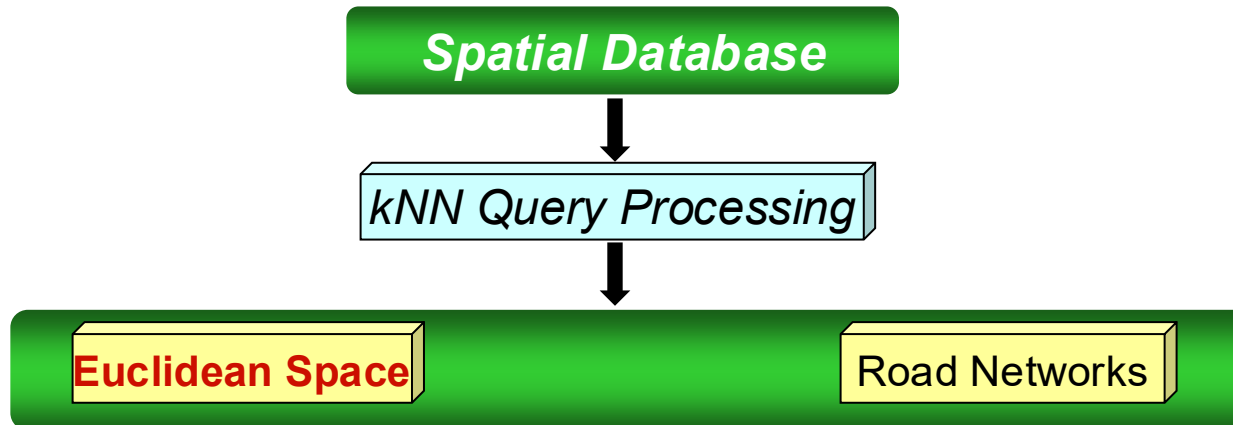
- ✓ **NN Query:** Roussopoulos et al., SIGMOD 1995
- ✓ **K-NN for moving query point:** Zong et al., SSTD 2001
- ✓ **Time-parameterized queries :** Tao et al., SIGMOD 2002
- ✓ **Continuous NN Search:** Tao et al., VLDB 2002

Object Based

Related Work



Related Work

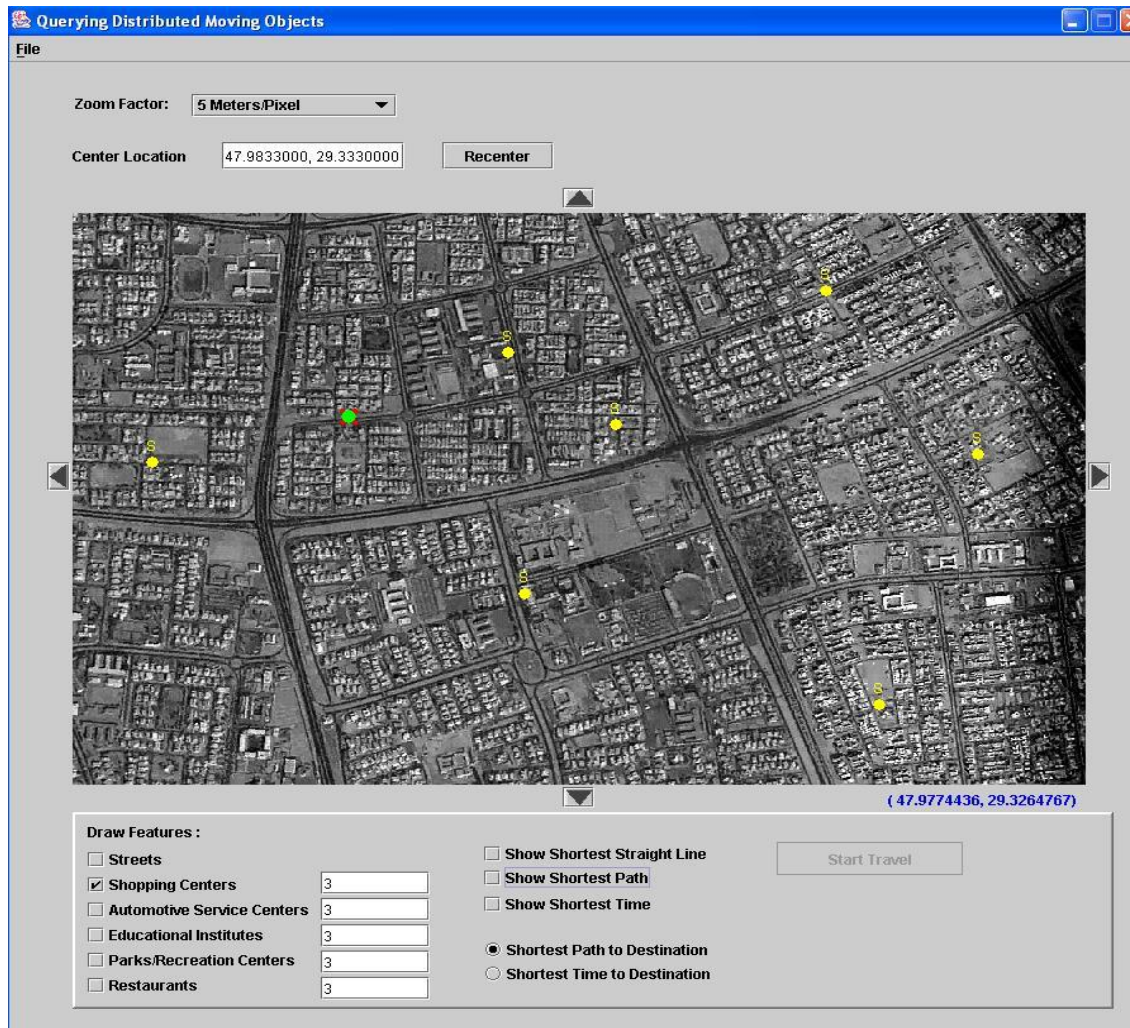


- ✓ Sina-Continuous querying: Mokbel et al., SIGMOD 2004
- ✓ Sea-CNN queries: Xiong et al., ICDE 2005
- ✓ Monitoring kNN on moving objects: Yu et al., ICDE 2005
- ✓ Conceptual partitioning: Mouratidis et al., SIGMOD 2005

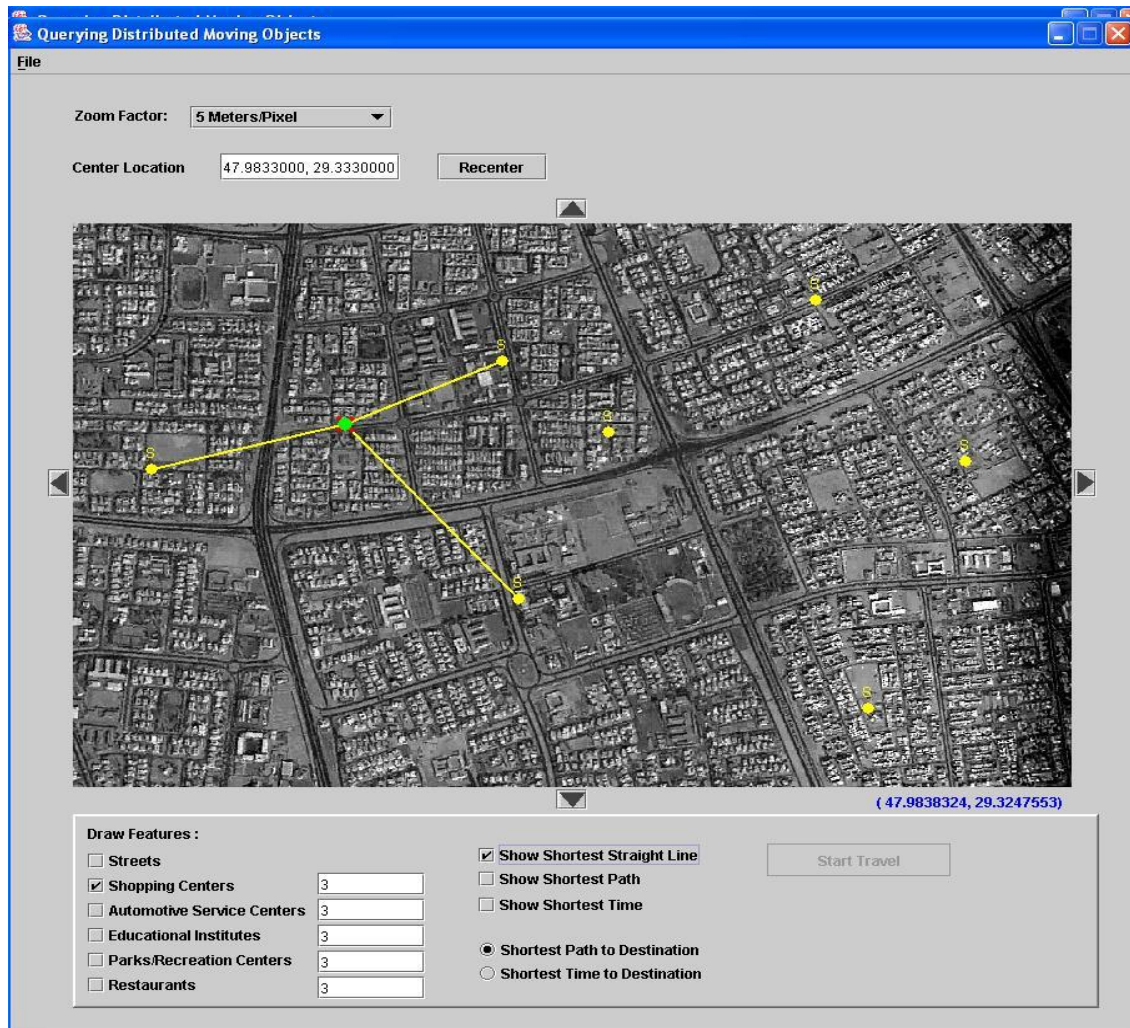
Space Based

Example 1:

Finding the 3 closest shopping centers

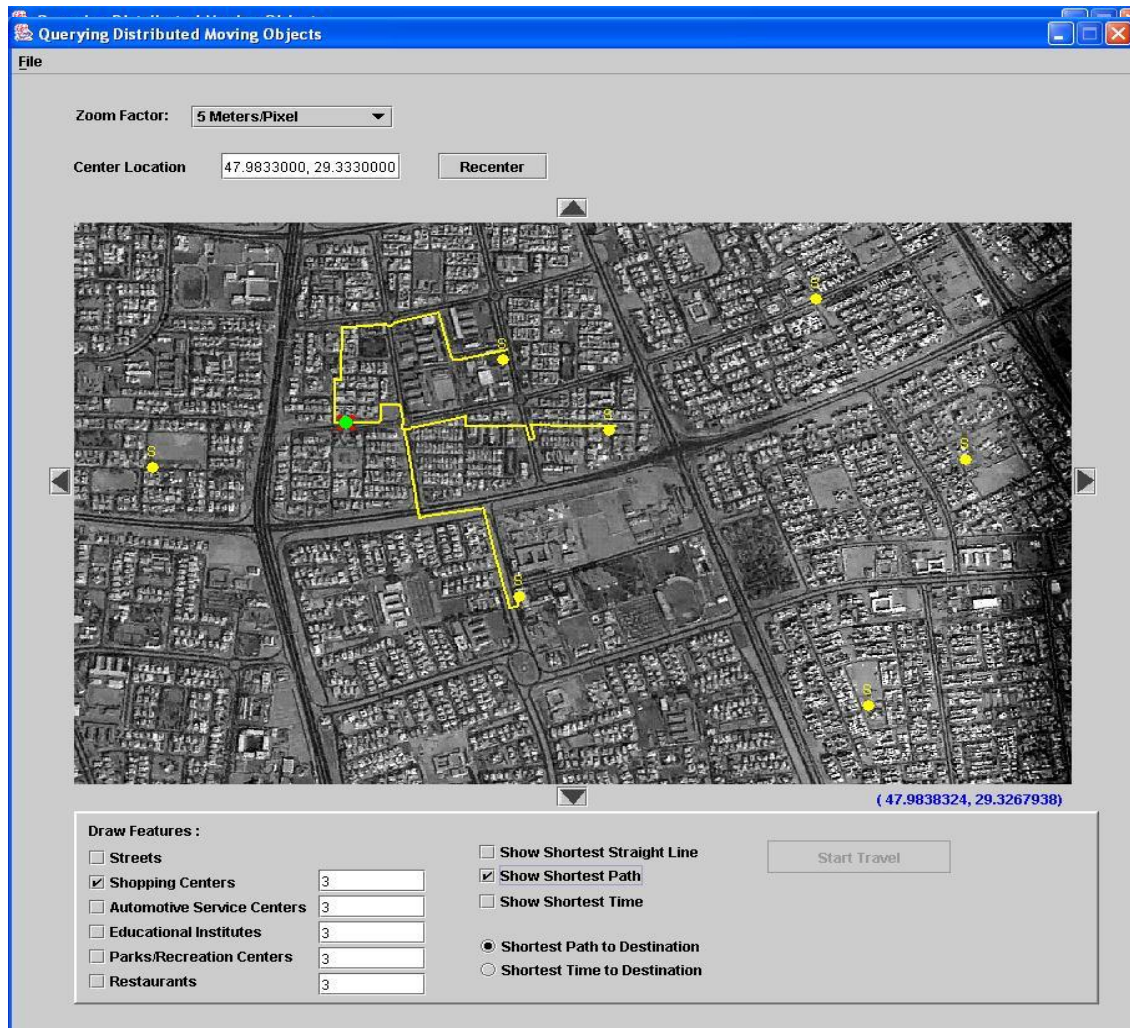


Example 1: Finding the 3 closest shopping centers



Example 1:

Finding the 3 closest shopping centers



Example 2:

Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Yellow Pages - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail

Address [McClintock+Ave&city=Los+Angeles&state=CA&zip=90089-0063&country=us&msa=44808&sl=34.019413&sl=-118.290486&cs=9](#) Go Links

YAHOO! GetLocal Yellow Pages [Yahoo! - Help](#)

Three Ways to Get Local: [Maps](#) [Yellow Pages](#) [City Guides](#)

CUBE CLUB tour 2002 **MUSIC + GAMING + TECHNOLOGY**

Welcome, Guest User [Create My Favorite Locations - Sign In](#)

Yahoo! Yellow Pages

3740 McClintock Ave, Los Angeles CA [Change Location](#) - [Save This Location](#)

[Top](#) > [Food and Dining](#) > [Restaurants](#) > All Restaurants

Sponsored Businesses | [All Businesses](#) Showing 1-3 of 3 | [Previous](#) | [Next](#)

Business Name	Address	Miles**
Mrs. Bethune's Tea Cakes (323) 292-0400 Website	5445 10th Avenue Los Angeles, CA Map	2.8
Vegetarian Affair (323) 292-8664 Website	3715 Santa Rosalia Drive Los Angeles, CA Map	2.8
Tropic Caribbean Flava (323) 294-8096 Website	3835 Santa Rosalia Drive Los Angeles, CA Map	2.9

All Businesses Showing 1-17 of 78 | [Previous](#) | [Next](#)

Internet

Example 2:

Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Yellow Pages - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail

Address Go Links

YAHOO! GetLocal Yellow Pages  [Yahoo! - Help](#)

Three Ways to Get Local:  [Maps](#)  [Yellow Pages](#)  [City Guides](#)

CUBE CLUB tour 2002 **MUSIC + GAMING + TECHNOLOGY**

Welcome, Guest User [Create My Favorite Locations - Sign In](#)

Yahoo! Yellow Pages

3740 Mcclintock Ave, Los Angeles CA [Change Location](#) - [Save This Location](#)

[Top](#) > [Food and Dining](#) > [Restaurants](#) > All Restaurants

Sponsored Businesses | [All Businesses](#) Showing 1-3 of 3 | P

Business Name	Address	
Mrs. Bethune's Tea Cakes (323) 292-0400 Website	5445 10th Avenue Los Angeles, CA Map	2.8
Vegetarian Affair (323) 292-8664 Website	3715 Santa Rosalia Drive Los Angeles, CA Map	2.8
Tropic Caribbean Flava (323) 294-8096 Website	3835 Santa Rosalia Drive Los Angeles, CA Map	2.9

All Businesses Showing 1-17 of 78 | P

Internet

Example 2: Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 5445 10th Avenue, Los Angeles, CA 90043-2525 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Mail Print Mailbox Yahoo! Messenger

Address [=Use+Address+Below&newtaddr=5445+10th+Avenue&newtcsc=los+angeles%2C+ca&newtcountry=us&Submit=Get+Directions](#) Go Links >>

Yahoo! maps [Maps Home](#)

[Maps](#) | [Driving Directions](#)

Starting from: ① 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: ② 5445 10th Avenue, Los Angeles, CA 90043-2525 [Save Address](#) [Get Reverse Directions](#)

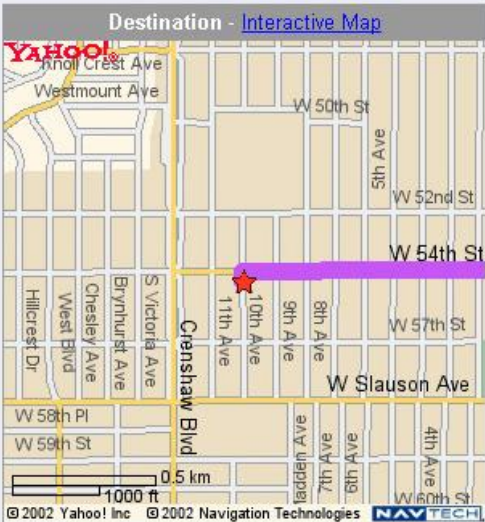
Distance: 4.4 miles **Approximate Travel Time:** 10 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

Full Route



Destination - Interactive Map



© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Done Internet

Example 2: Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 5445 10th Avenue, Los Angeles, CA 90043-2525 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail News RSS Yahoo! Messenger

Address [=Use+Address+Below&newtaddr=5445+10th+Avenue&newtcscz=los+angeles%2C+ca&newtcountry=us&Submit=Get+Directions](#) Go Links >>

Yahoo! maps [Maps Home](#)

[Maps](#) | [Driving Directions](#)

Starting from: ① 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: ② 5445 10th Avenue, Los Angeles, CA 90043-2525 [Save Address](#) [Get Reverse Directions](#)

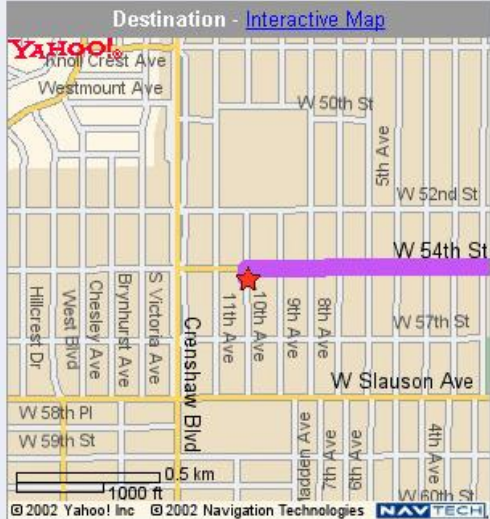
Distance: **4.4 miles** Estimated Travel Time: 10 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

Full Route



Destination - Interactive Map



© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Done Internet

Example 2: Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 3715 Santa Rosalia Drive, Los Angeles, CA 90008-3603 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail

Address Address+Below&newtdr=3715+Santa+Rosalia+Drive&newtcz=los+angeles%2C+ca&newcountry=us&Submit=Get+Directions Go Links

Maps | Driving Directions

Starting from: 1 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: 2 3715 Santa Rosalia Drive, Los Angeles, CA 90008-3603 [Save Address](#) [Get Reverse Directions](#)

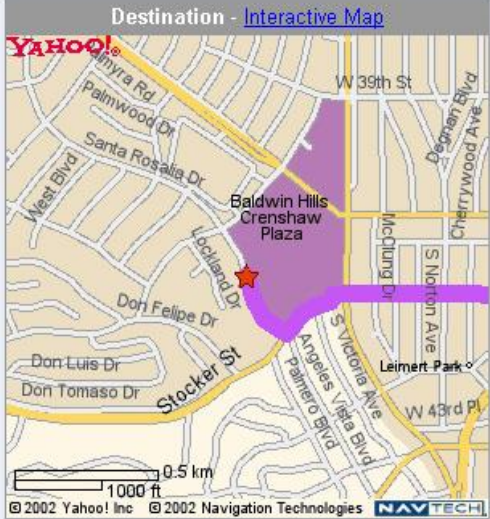
Distance: 3.9 miles Approximate Travel Time: 9 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

Full Route



Destination - Interactive Map



Done Internet

Example 2: Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 3715 Santa Rosalia Drive, Los Angeles, CA 90008-3603 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail News Groups

Address Address+Below&newaddr=3715+Santa+Rosalia+Drive&newtcz=los+angeles%2C+ca&newcountry=us&Submit=Get+Directions Go Links

Maps | Driving Directions

Starting from: 1 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: 2 3715 Santa Rosalia Drive, Los Angeles, CA 90008-3603 [Save Address](#) [Get Reverse Directions](#)

Distance: 3.9 miles proximate Travel Time: 9 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

Full Route

Destination - Interactive Map

© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Example 2:

Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 3835 Santa Rosalia Drive, Los Angeles, CA 90008-2540 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail

Address Address+Below&newaddr=3835+Santa+Rosalia+Drive&newtcz=los+angeles%2C+ca&newcountry=us&Submit=Get+Directions Go Links

Yahoo! Maps

[Maps Home](#)

[Maps](#) | [Driving Directions](#)

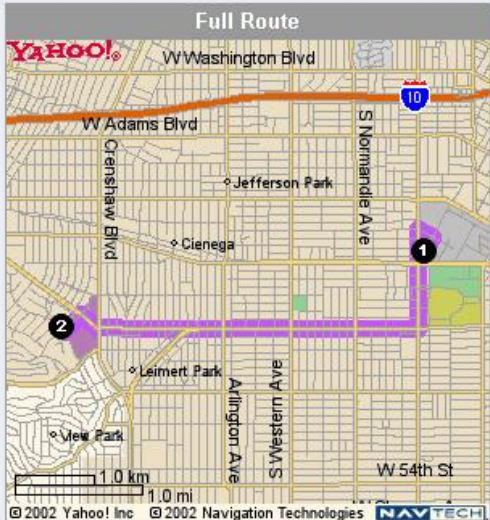
Starting from: ① 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: ② 3835 Santa Rosalia Drive, Los Angeles, CA 90008-2540 [Save Address](#) [Get Reverse Directions](#)

Distance: 3.9 miles **Approximate Travel Time:** 9 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

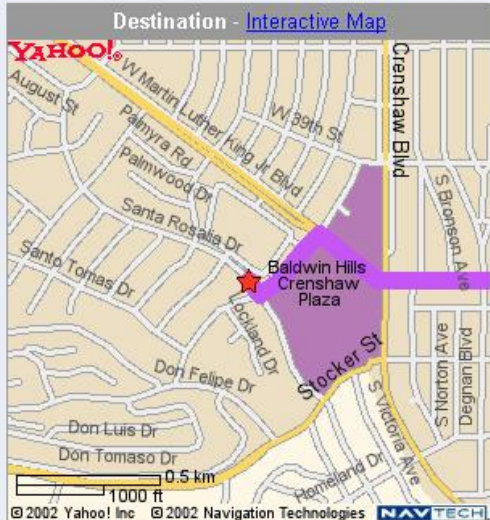
Full Route



W Washington Blvd
W Adams Blvd
Crenshaw Blvd
Jefferson Park
Cienega
S Normandie Ave
Leimert Park
Arlington Ave
W 54th St
W 56th St
W 58th St
W 60th St
W 62nd St
W 64th St
W 66th St
W 68th St
W 70th St
W 72nd St
W 74th St
W 76th St
W 78th St
W 80th St
W 82nd St
W 84th St
W 86th St
W 88th St
W 90th St
W 92nd St
W 94th St
W 96th St
W 98th St
W 100th St

© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Destination - Interactive Map



W Martin Luther King Jr Blvd
W 39th St
Crenshaw Blvd
S Bronson Ave
S Norton Ave
Degenan Blvd
W 38th St
W 37th St
W 36th St
W 35th St
W 34th St
W 33rd St
W 32nd St
W 31st St
W 30th St
W 29th St
W 28th St
W 27th St
W 26th St
W 25th St
W 24th St
W 23rd St
W 22nd St
W 21st St
W 20th St
W 19th St
W 18th St
W 17th St
W 16th St
W 15th St
W 14th St
W 13th St
W 12th St
W 11th St
W 10th St
W 9th St
W 8th St
W 7th St
W 6th St
W 5th St
W 4th St
W 3rd St
W 2nd St
W 1st St
W 0th St

© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Example 2: Finding the 3 closest restaurants to USC with Yahoo

Yahoo! Maps - Directions to 3835 Santa Rosalia Drive, Los Angeles, CA 90008-2540 - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Media Print Mail Yahoo! Messenger

Address Address+Below&newaddr=3835+Santa+Rosalia+Drive&newtcz=los+angeles%2C+ca&newcountry=us&Submit=Get+Directions Go Links

Yahoo! Maps

[Maps Home](#)

[Maps](#) | [Driving Directions](#)

Starting from: ① 3740 McClintock Ave, Los Angeles, CA 90089-0063 [Save Address](#)

Arriving at: ② 3835 Santa Rosalia Drive, Los Angeles, CA 90008-2540 [Save Address](#) [Get Reverse Directions](#)

Distance: 3.9 miles **Approximate Travel Time:** 9 mins

[Email Directions](#) [Printable Version](#) [Text Only Driving Directions](#)

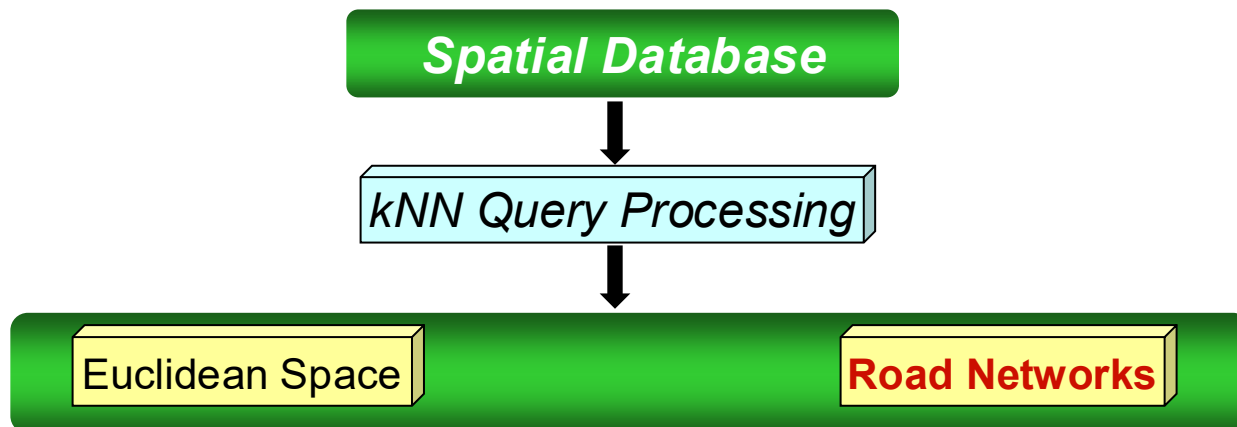
Full Route

© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Destination - Interactive Map

© 2002 Yahoo! Inc © 2002 Navigation Technologies NAVTECH

Related Work



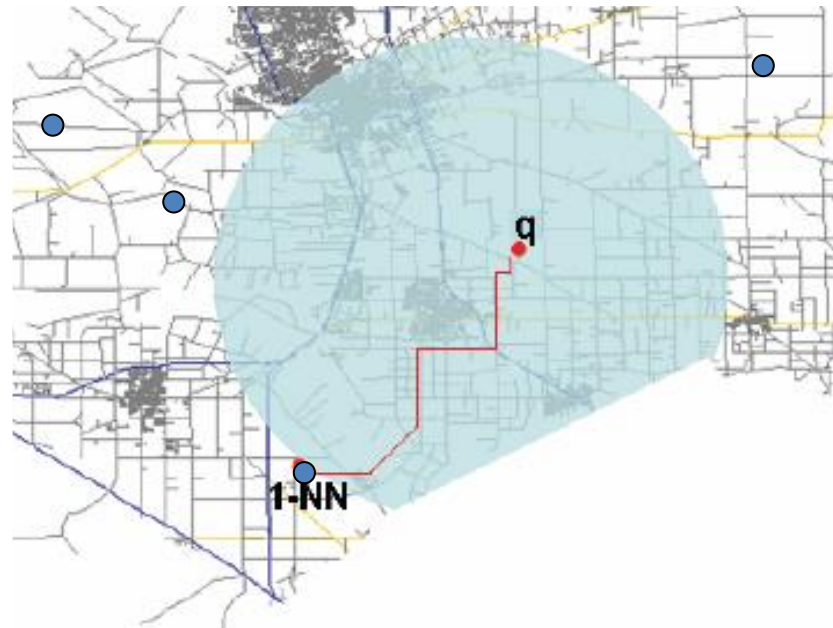
- ✓ **Road Network Embedding for K-NN Search: Shahabi et al., ACM-GIS 2002**
- ✓ **Query processing in SNDB : Papadias et al., VLDB 2003**
- ✓ **Voronoi-based kNN in SNDB: Kolahdouzan et al., VLDB 2004**

Related Work

Query processing in SNDB : Papadias et al., VLDB 2003

Incremental Network Expansion (INE)

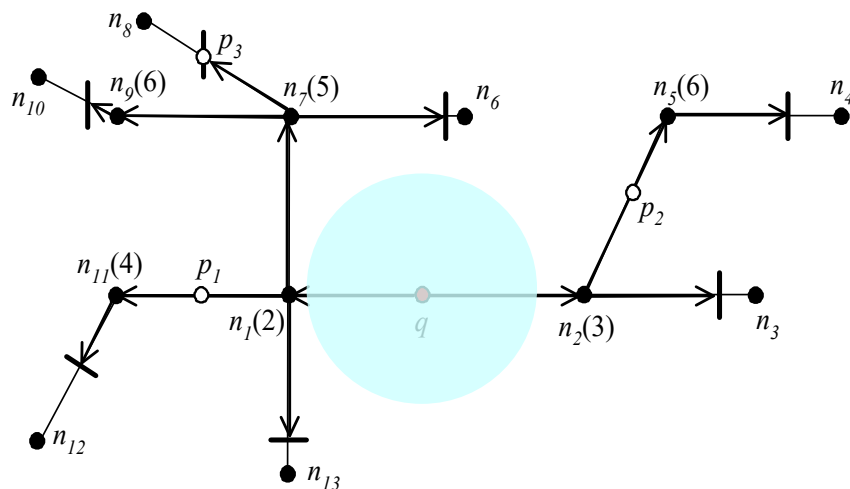
- Blind expansion hence redundant node access



Related Work

Query processing in SNDB : Papadias et al., VLDB 2003

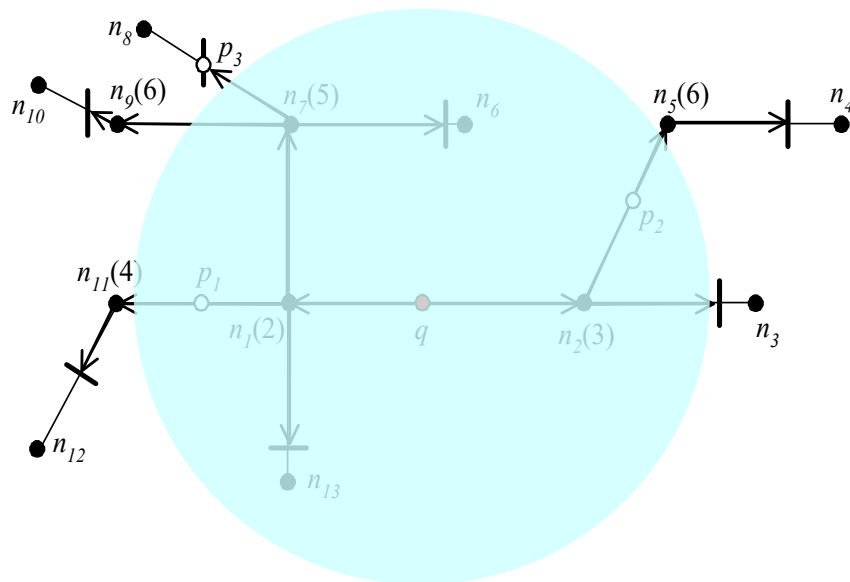
– Incremental Network Expansion



Related Work

Query processing in SNDB : Papadias et al., VLDB 2003

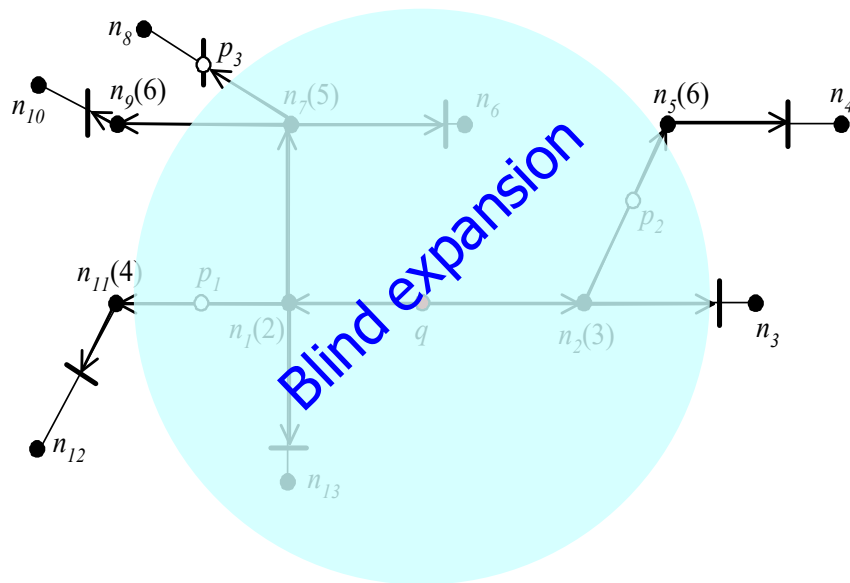
- Incremental Network Expansion



Related Work

Query processing in SNDB : Papadias et al., VLDB 2003

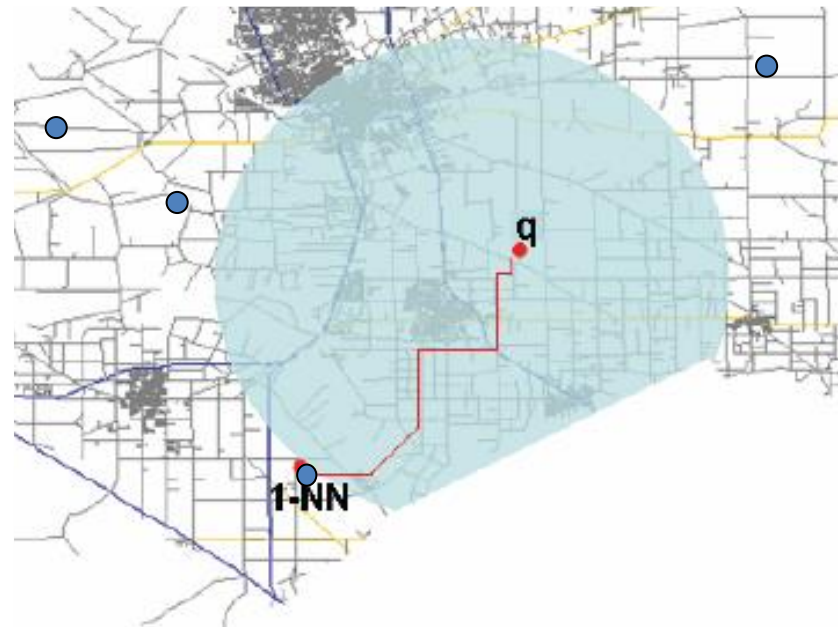
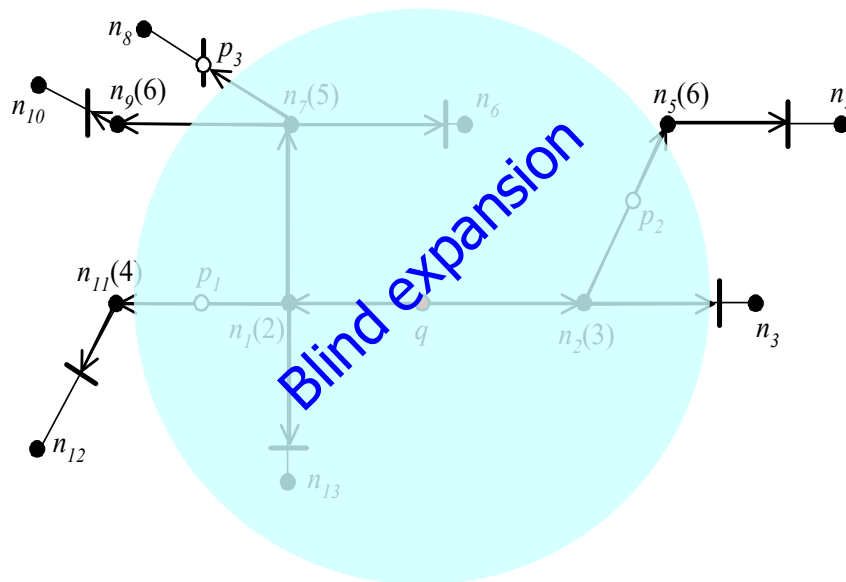
- Incremental Network Expansion



Related Work

Query processing in SNDB : Papadias et al., VLDB 2003

- Incremental Network Expansion



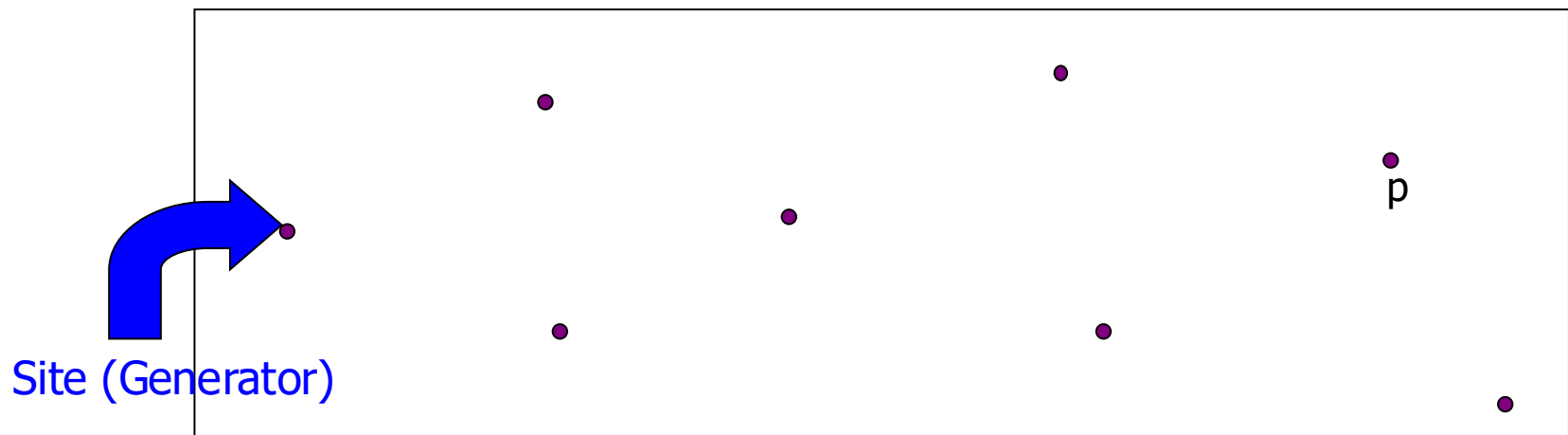
Preliminaries: Voronoi Diagram

- Given set of sites (POI), a Voronoi diagram partitions the plane into disjoint Voronoi polygons (cells), one for each site



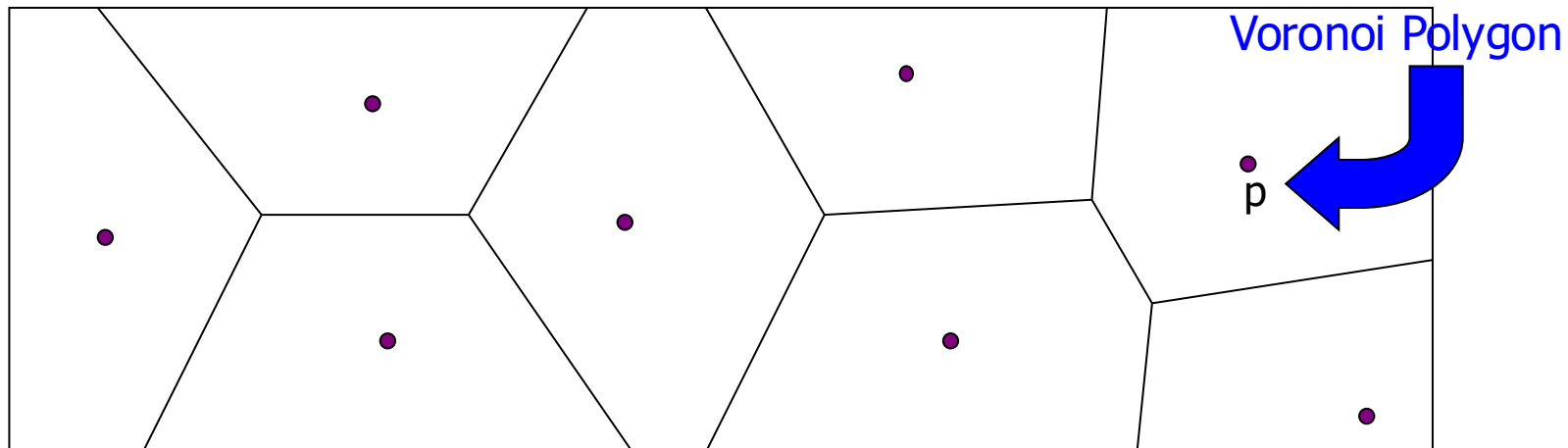
Preliminaries: Voronoi Diagram

- Given set of sites (POI), a Voronoi diagram partitions the plane into disjoint Voronoi polygons (cells), one for each site



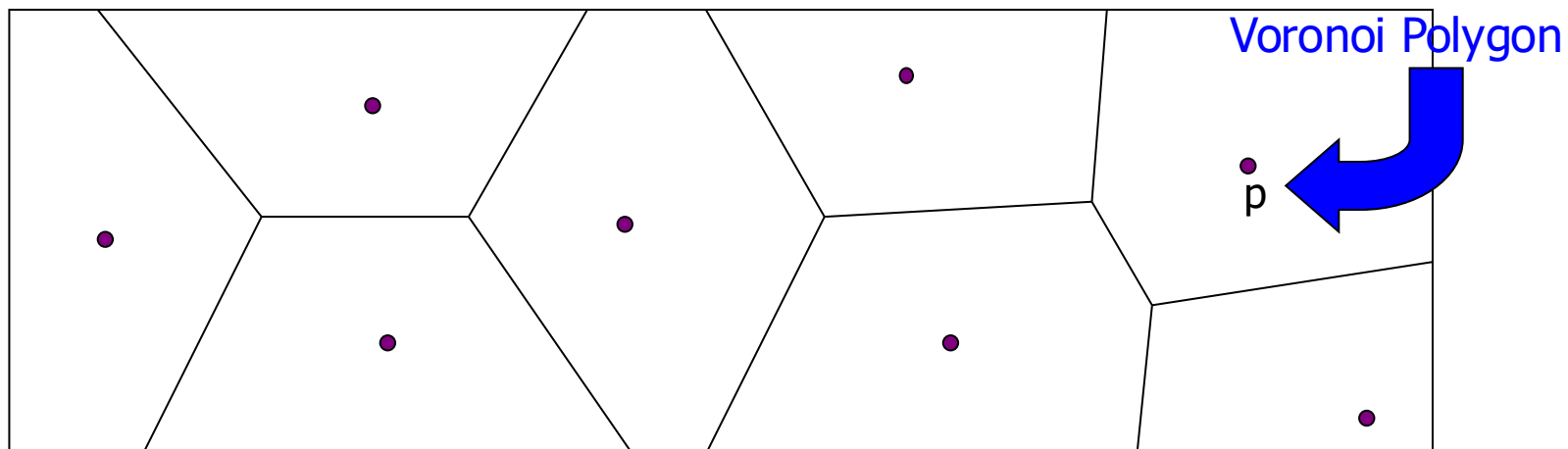
Preliminaries: Voronoi Diagram

- Given set of sites (POI), a Voronoi diagram partitions the plane into disjoint Voronoi polygons (cells), one for each site



Preliminaries: Voronoi Diagram

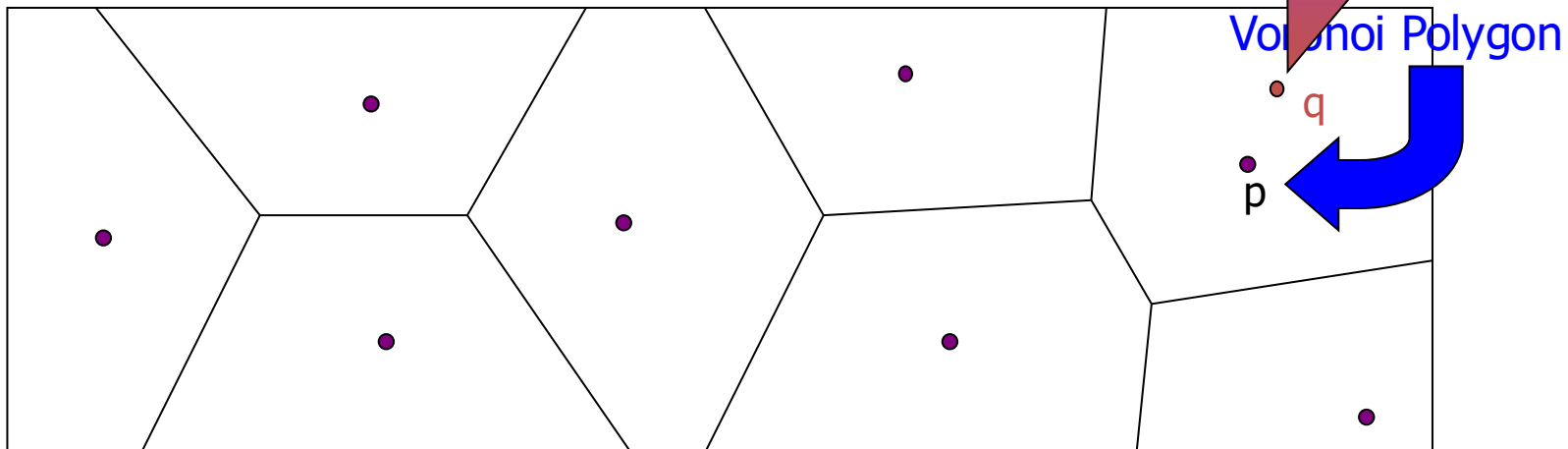
- Given set of sites (POI), a Voronoi diagram partitions the plane into disjoint Voronoi polygons (cells), one for each site
- The region including a site p includes all locations which are closer to p than to any other object p' .



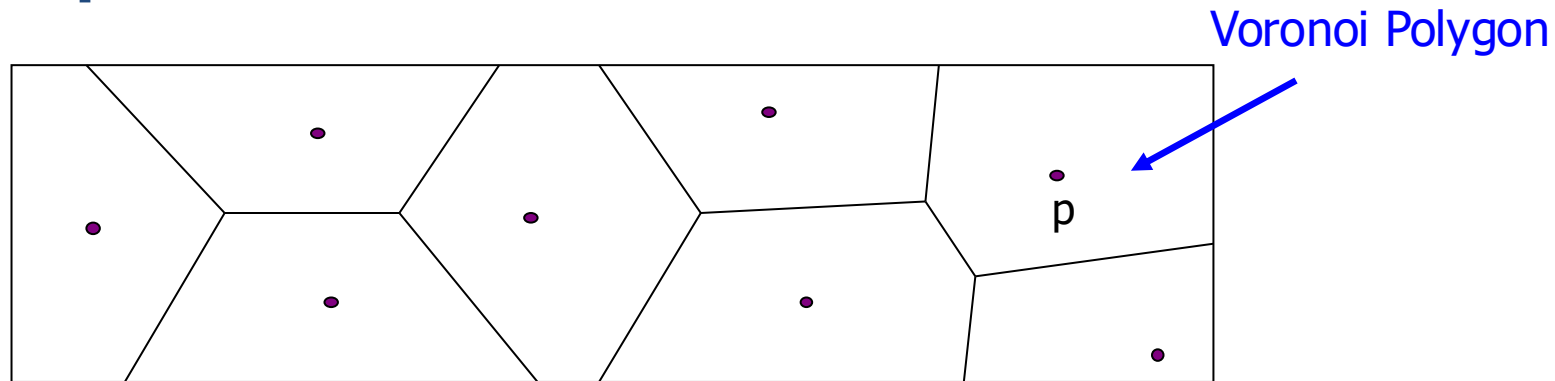
Preliminaries: Voronoi Diagram

- Given set of sites (POI), a Voronoi diagram partitions the plane into disjoint Voronoi polygons (each polygon contains one site)
- The region including a site p includes all points that are closer to p than to any other object p' .

q is closer to the generator of the Voronoi Polygon than any other generator

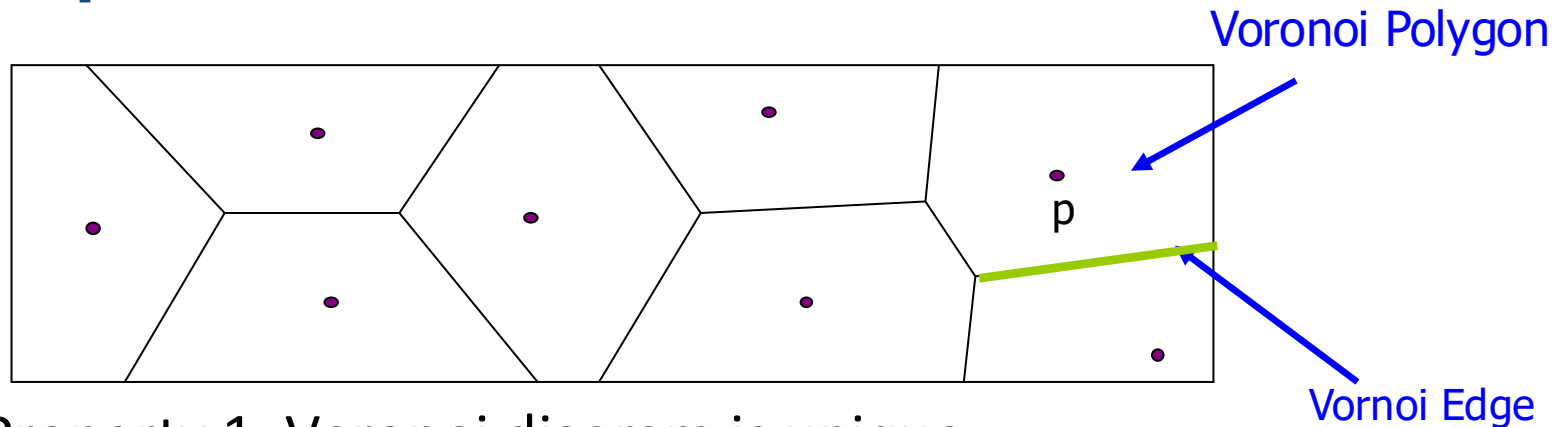


Preliminaries: Voronoi Properties



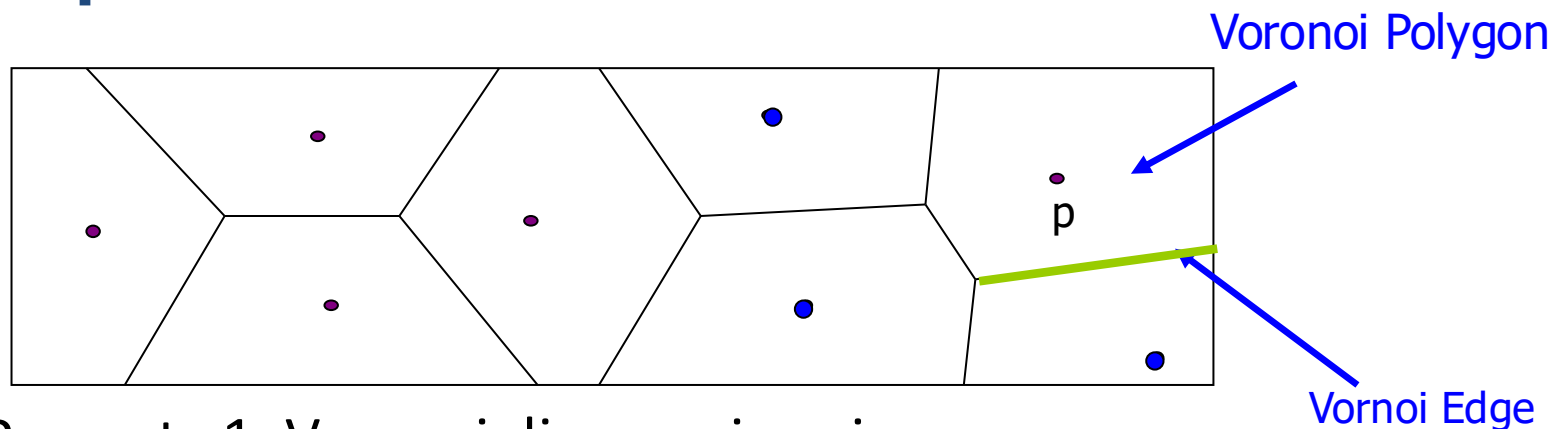
- Property 1: Voronoi diagram is unique

Preliminaries: Voronoi Properties



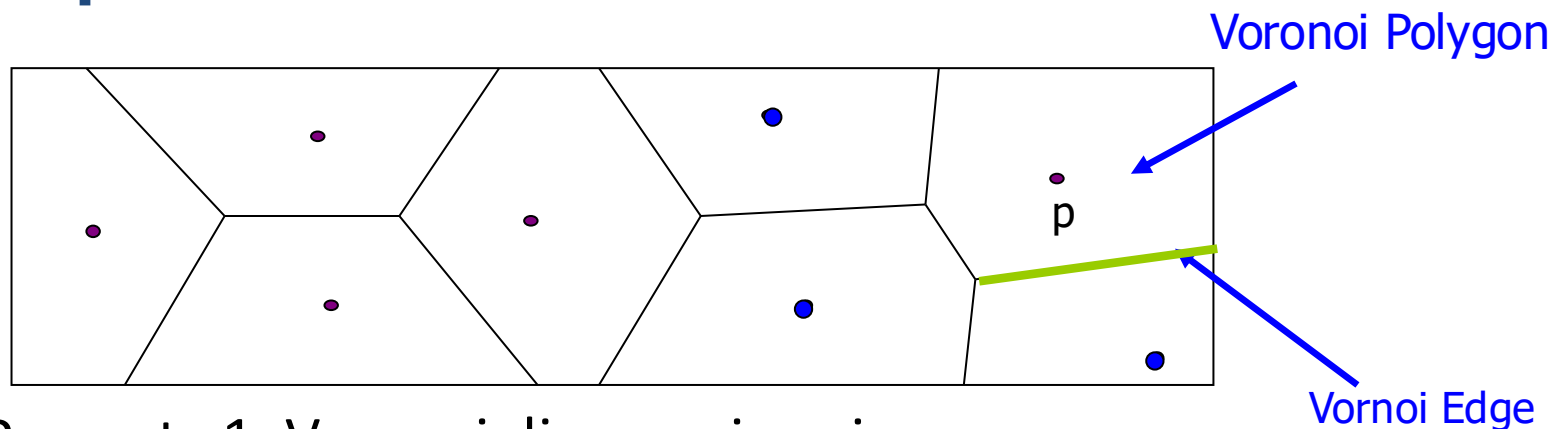
- Property 1: Voronoi diagram is unique
- Property 2: Voronoi edges are
 - shared by two generators
 - in equal distance to neighboring generators

Preliminaries: Voronoi Properties



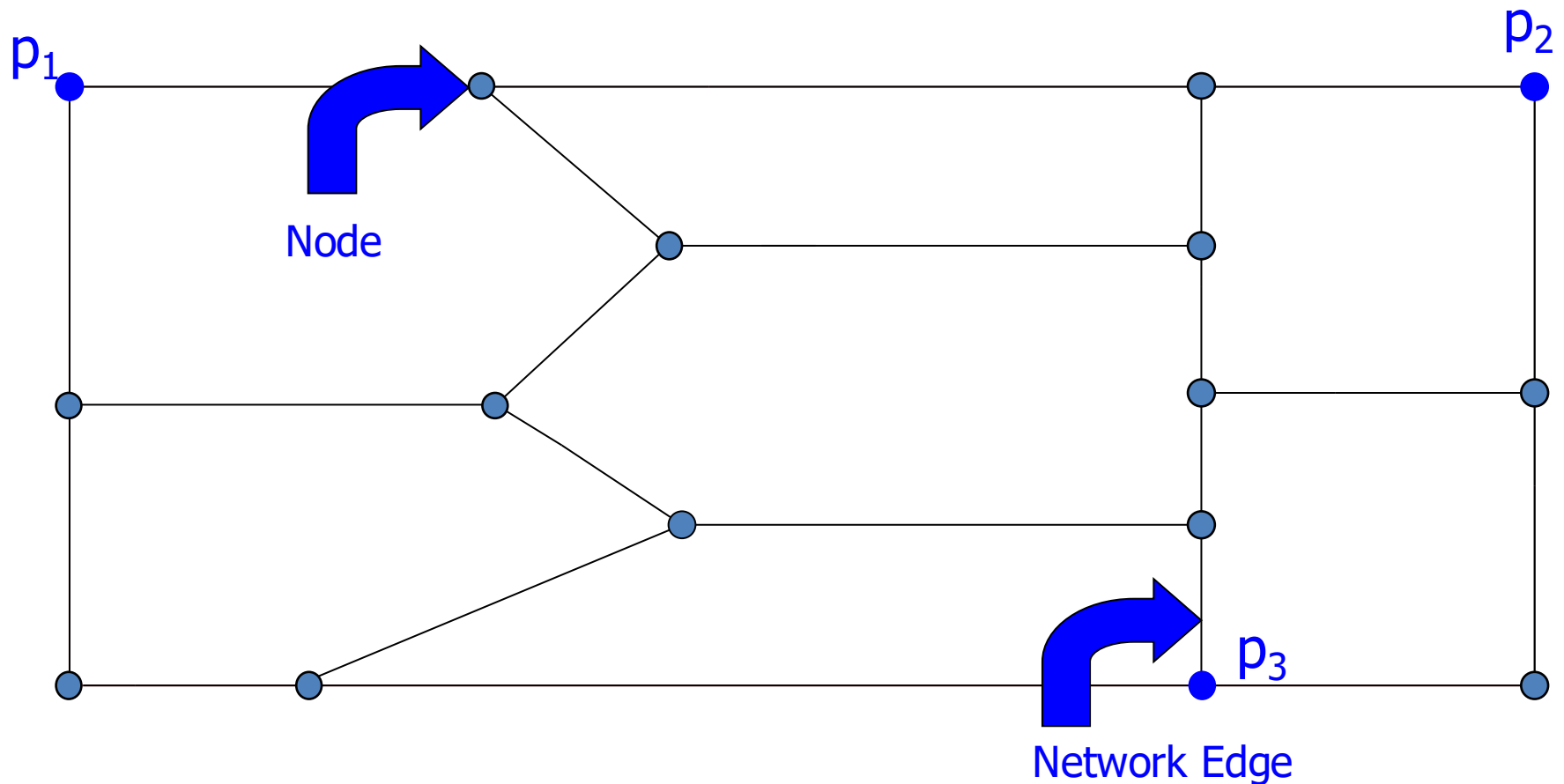
- Property 1: Voronoi diagram is unique
- Property 2: Voronoi edges are
 - shared by two generators
 - in equal distance to neighboring generators
- Property 3: Nearest generator of p is among the generators whose VP shares similar edges with p

Preliminaries: Voronoi Properties

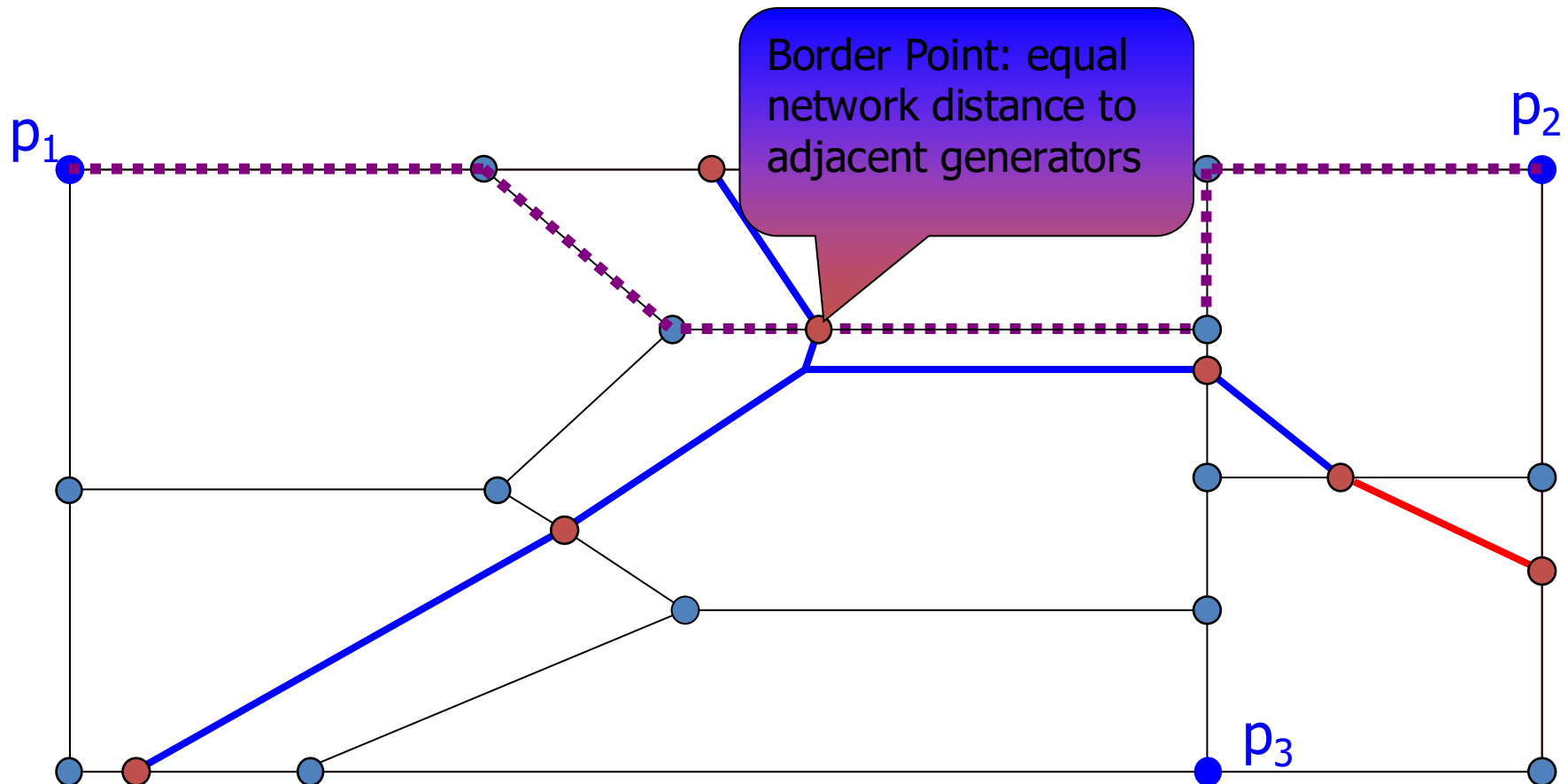


- Property 1: Voronoi diagram is unique
- Property 2: Voronoi edges are
 - shared by two generators
 - in equal distance to neighboring generators
- Property 3: Nearest generator of p is among the generators whose VP shares similar edges with p
- Property 4: Average number of Voronoi edges per VP is at most 6

Network Voronoi Diagram



Network Voronoi Diagram

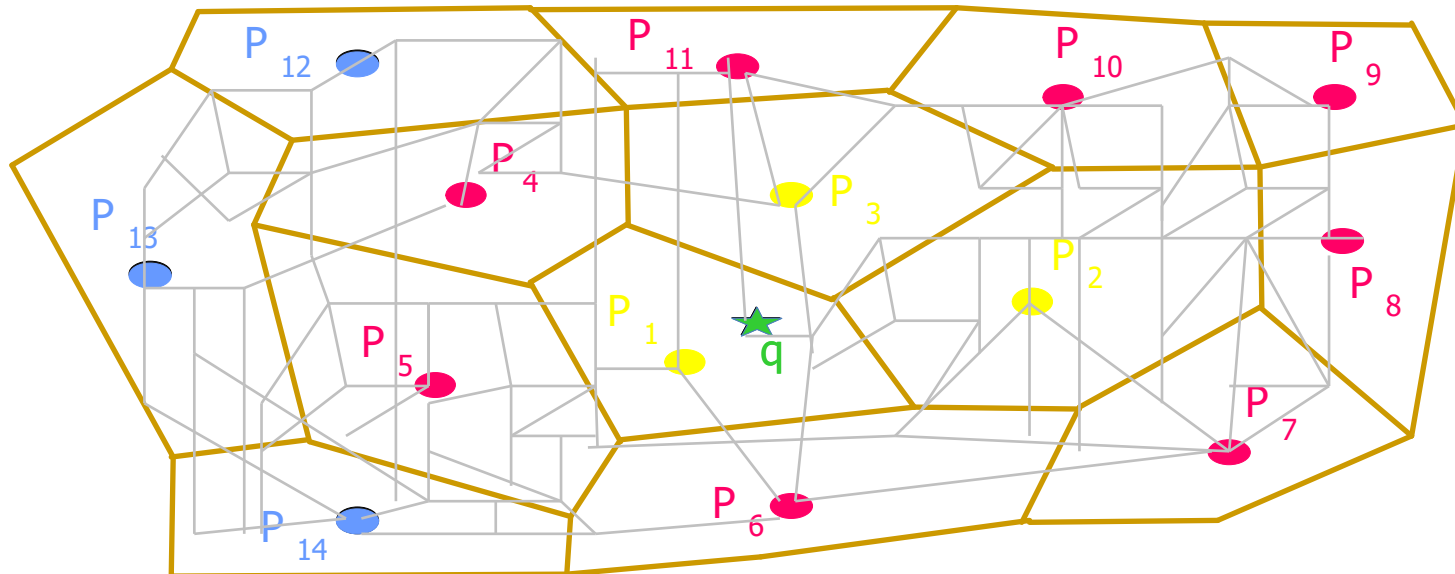


VN³ Approach

- Offline Index Generation
 1. Network Voronoi Construction
 2. Index Generation (R-tree)
 3. Distance Precomputation
- Online Query Processing
 1. Find 1st NN
 2. Find k NN-> Filter & Refine

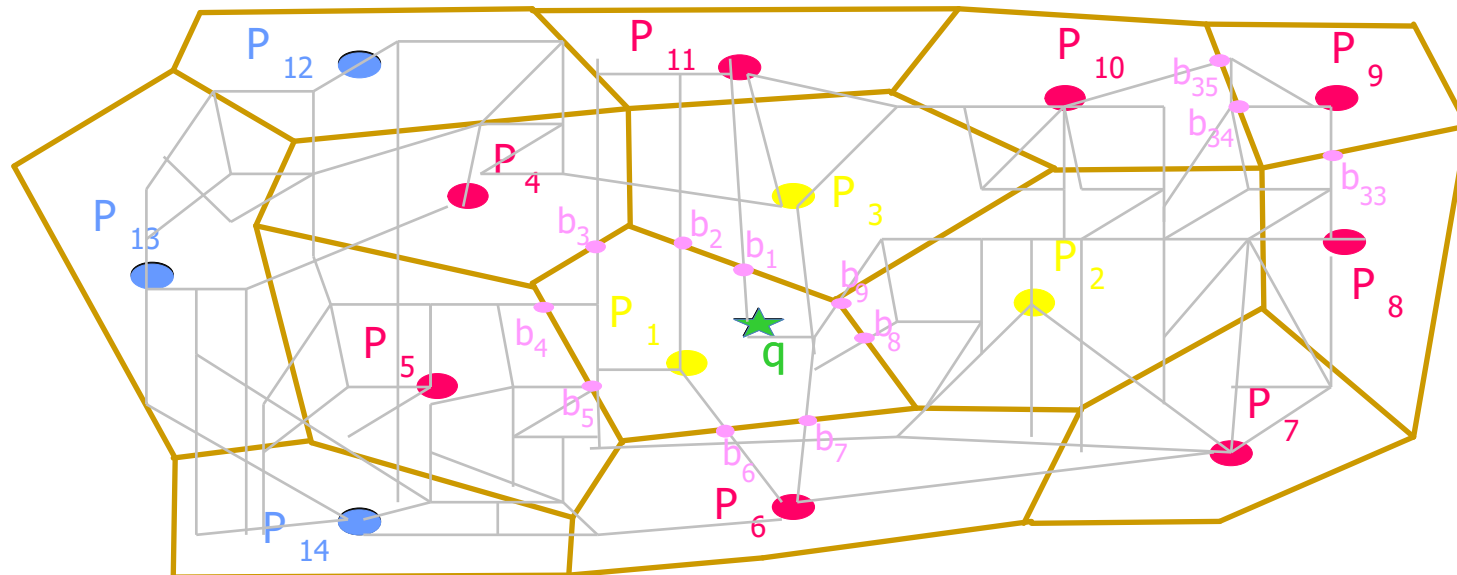
Overview

- How to compute distance from q to candidates?



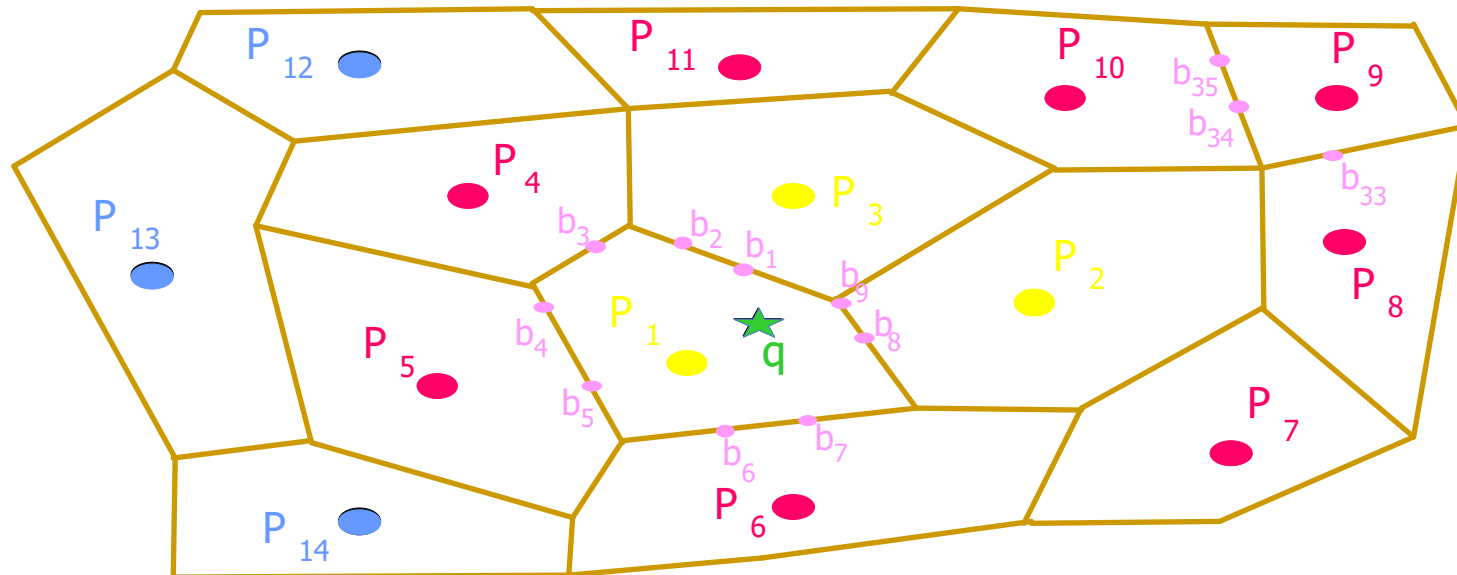
Overview

- How to compute distance from q to candidates?



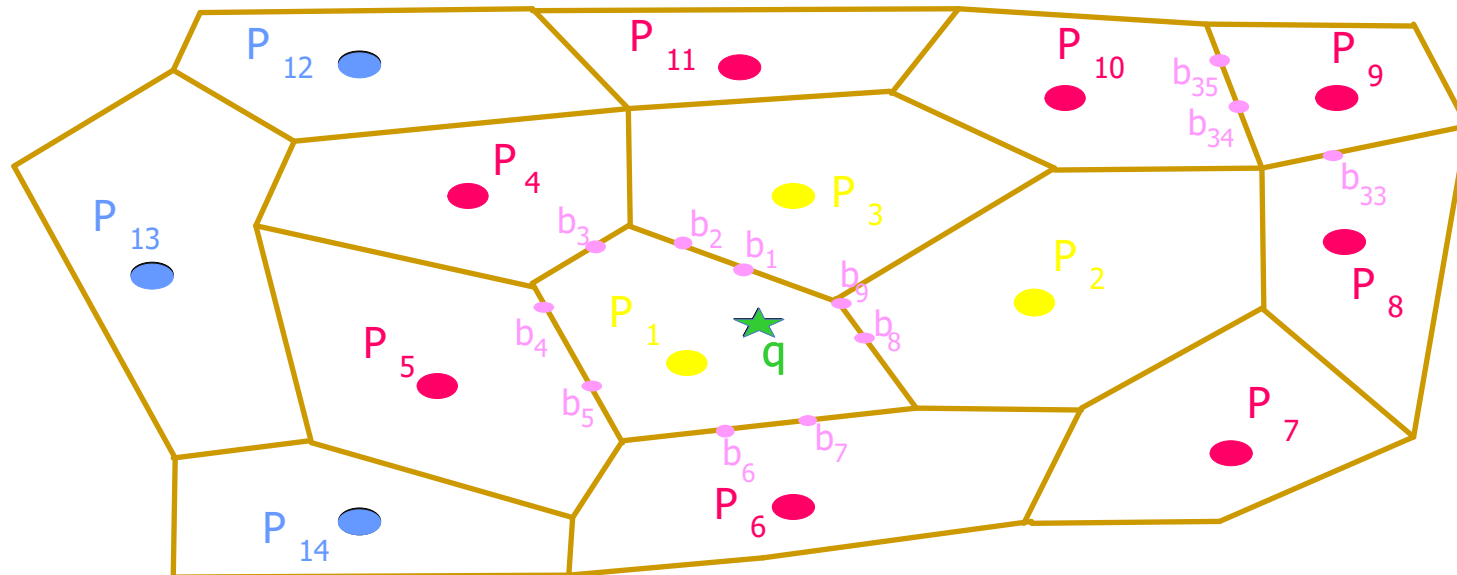
Overview

- How to compute distance from q to candidates?



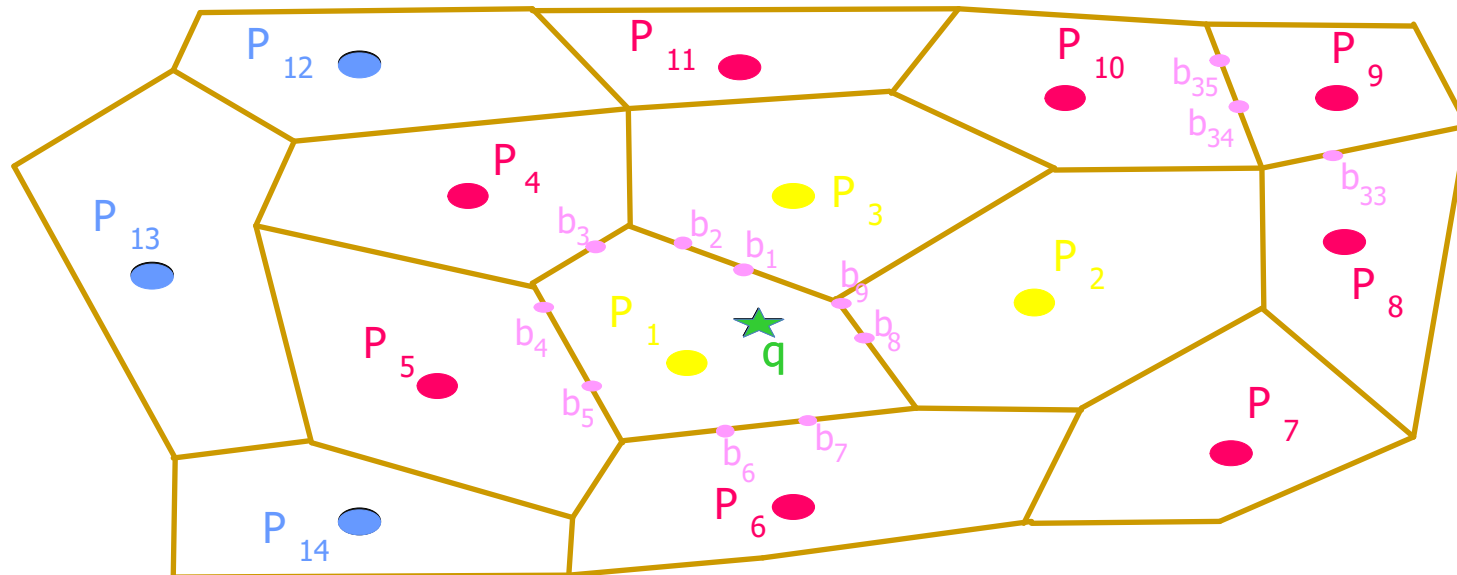
Overview

- How to compute distance from q to candidates?
- Break distance from q to any P_i to:
 1. Distance from q to its borders
 - I. Online Progressive Expansion
 - II. Offline Pre-calculation



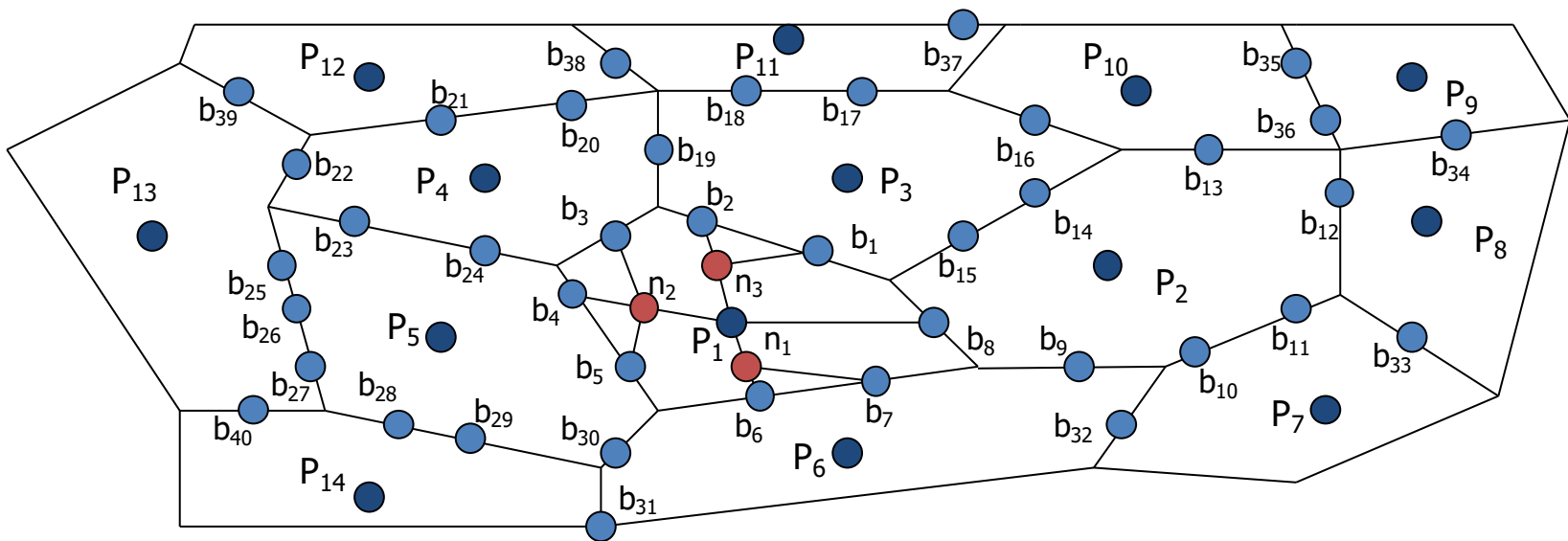
Overview

- How to compute distance from q to candidates?
- Break distance from q to any P_i to:
 1. Distance from q to its borders
 - I. Online Progressive Expansion
 - II. Offline Pre-calculation
 2. Distance from borders of q to borders of P_i
 - I. NVP Expansion
 - II. Distance Computing Optimization



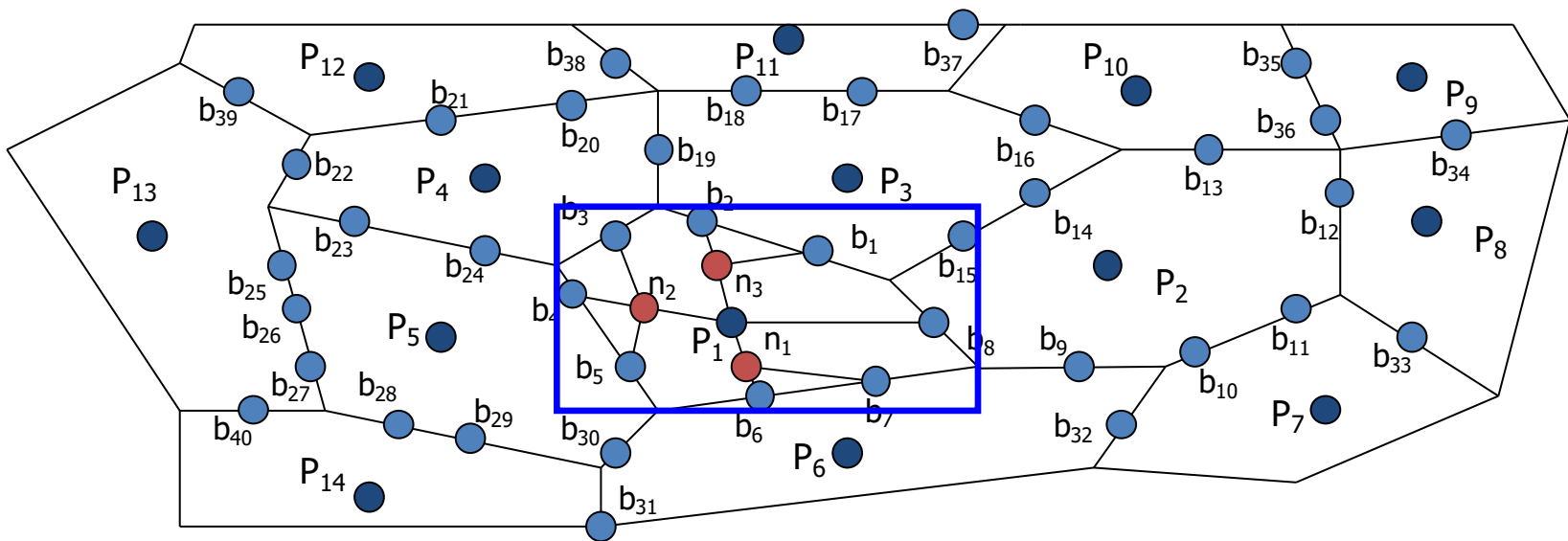
Offline Step

- Compute Network Voronoi Polygons (NVP)



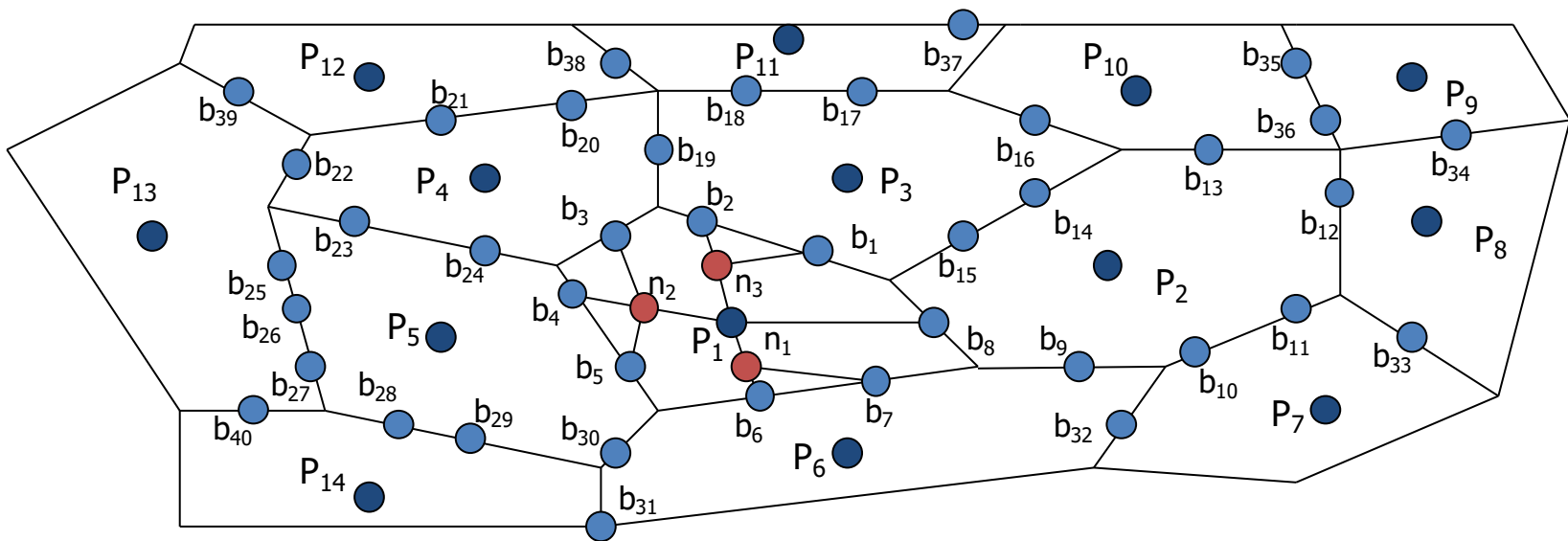
Offline Step

- Compute Network Voronoi Polygons (NVP)
- Index NVPs with R-Tree



Offline Step

- Compute Network Voronoi Polygons (NVP)
- Index NVPs with R-Tree
- Precompute the inter (across polygons) and intra (within polygons) network distances for each NVP

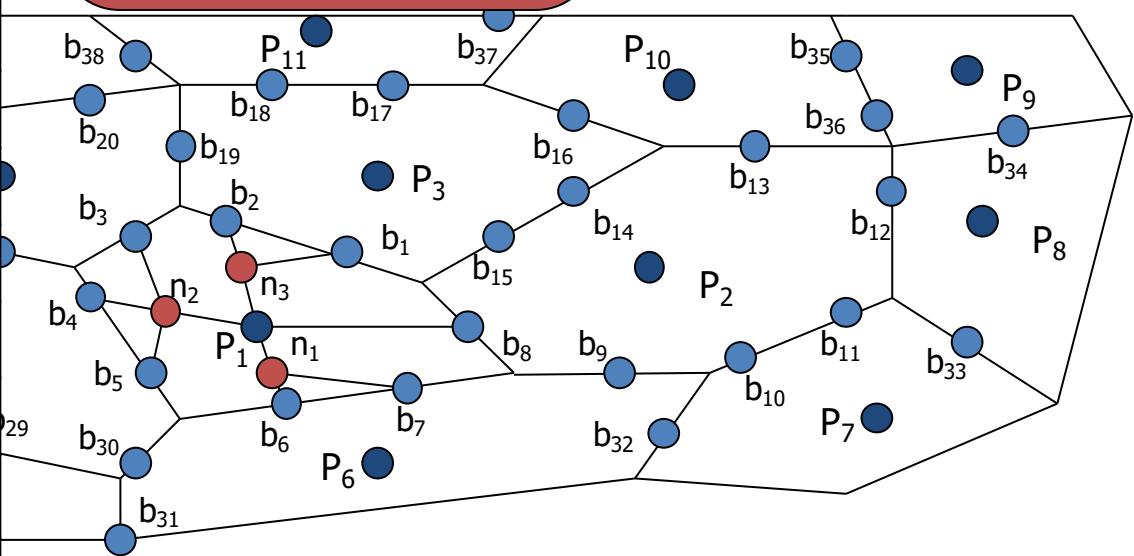


Offline Step

- Compute Network Voronoi Polygons (NVP)
- Index NVPs with
- Precompute the network distance

1. Precompute the Shortest Path distance from each generator to its border point
2. Precompute the Shortest Path between border points

b_{34}	P_9	$d_n(b_{34}, P_9)$
b_{35}	P_9	$d_n(b_{35}, P_9)$
b_{36}	P_9	$d_n(b_{36}, P_9)$
...	...	
b_1	b_2	$d_n(b_1, b_2)$
b_1	b_{15}	$d_n(b_1, b_{15})$
b_1	b_{13}	$d_n(b_1, b_{13})$
...	...	

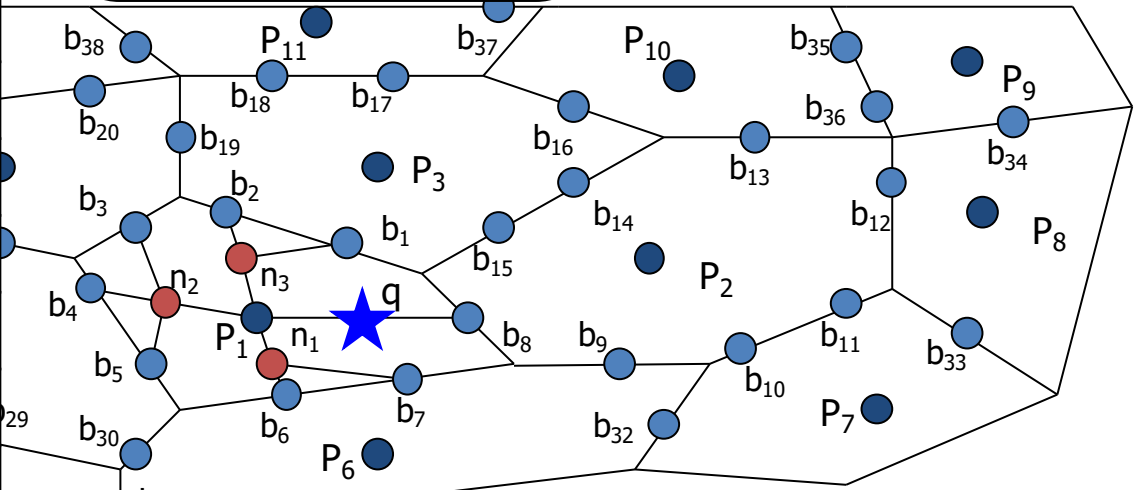


Offline Step

- Compute Network Voronoi Polygons (NVP)
- Index NVPs with
- Precompute the network distance

1. Precompute the Shortest Path distance from each generator to its border point
2. Precompute the Shortest Path between border points

b_{34}	P_9	$d_n(b_{34}, P_9)$
b_{35}	P_9	$d_n(b_{35}, P_9)$
b_{36}	P_9	$d_n(b_{36}, P_9)$
...	...	
b_1	b_2	$d_n(b_1, b_2)$
b_1	b_{15}	$d_n(b_1, b_{15})$
b_1	b_{13}	$d_n(b_1, b_{13})$
...	...	



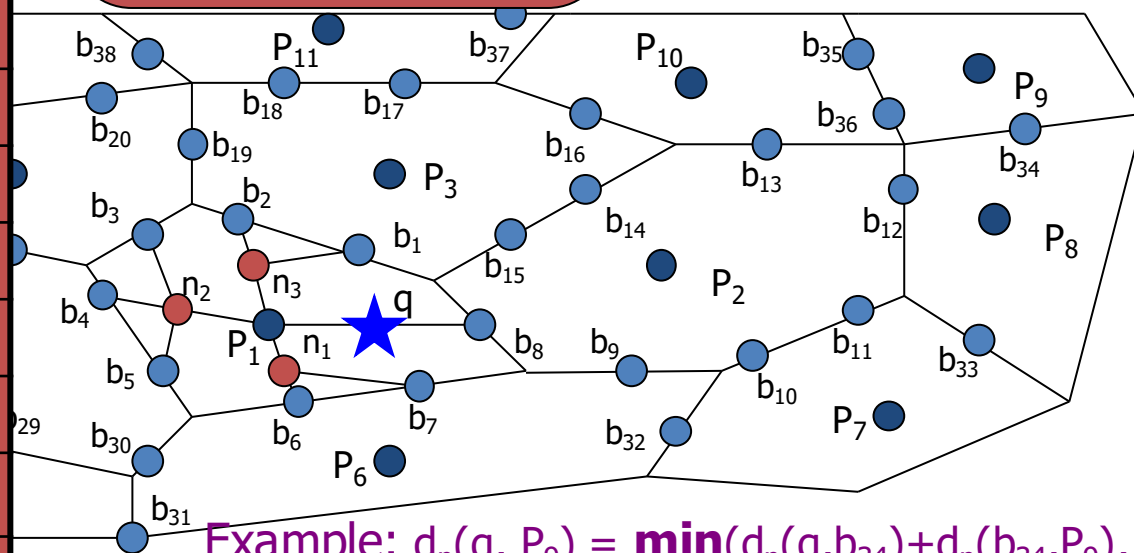
Example: $d_n(q, P_9) = \min(d_n(q, b_{34}) + d_n(b_{34}, P_9), d_n(q, b_{35}) + d_n(b_{35}, P_9), d_n(q, b_{36}) + d_n(b_{36}, P_9))$

Offline Step

- Compute Network Voronoi Polygons (NVP)
- Index NVPs with
- Precompute the network distance

1. Precompute the Shortest Path distance from each generator to its border point
2. Precompute the Shortest Path between border points

b_{34}	P_9	$d_n(b_{34}, P_9)$
b_{35}	P_9	$d_n(b_{35}, P_9)$
b_{36}	P_9	$d_n(b_{36}, P_9)$
...	...	
b_1	b_2	$d_n(b_1, b_2)$
b_1	b_{15}	$d_n(b_1, b_{15})$
b_1	b_{13}	$d_n(b_1, b_{13})$
...	...	

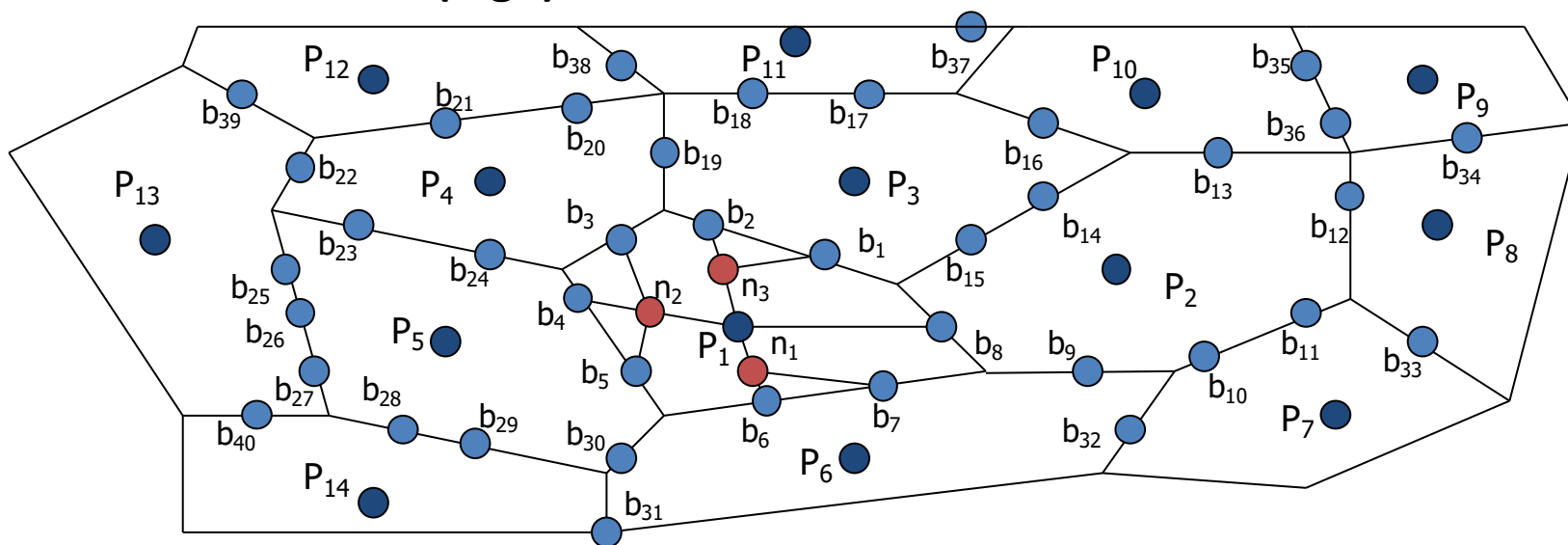


Example: $d_n(q, P_9) = \min(d_n(q, b_{34}) + d_n(b_{34}, P_9), d_n(q, b_{35}) + d_n(b_{35}, P_9), d_n(q, b_{36}) + d_n(b_{36}, P_9))$
 $d_n(q, b_{34}) = \min(d_n(q, b_1) + d_n(b_1, b_{34}), d_n(q, b_2) + d_n(b_2, b_{34}), \dots, d_n(q, b_8) + d_n(b_8, b_{34}))$

Online Step

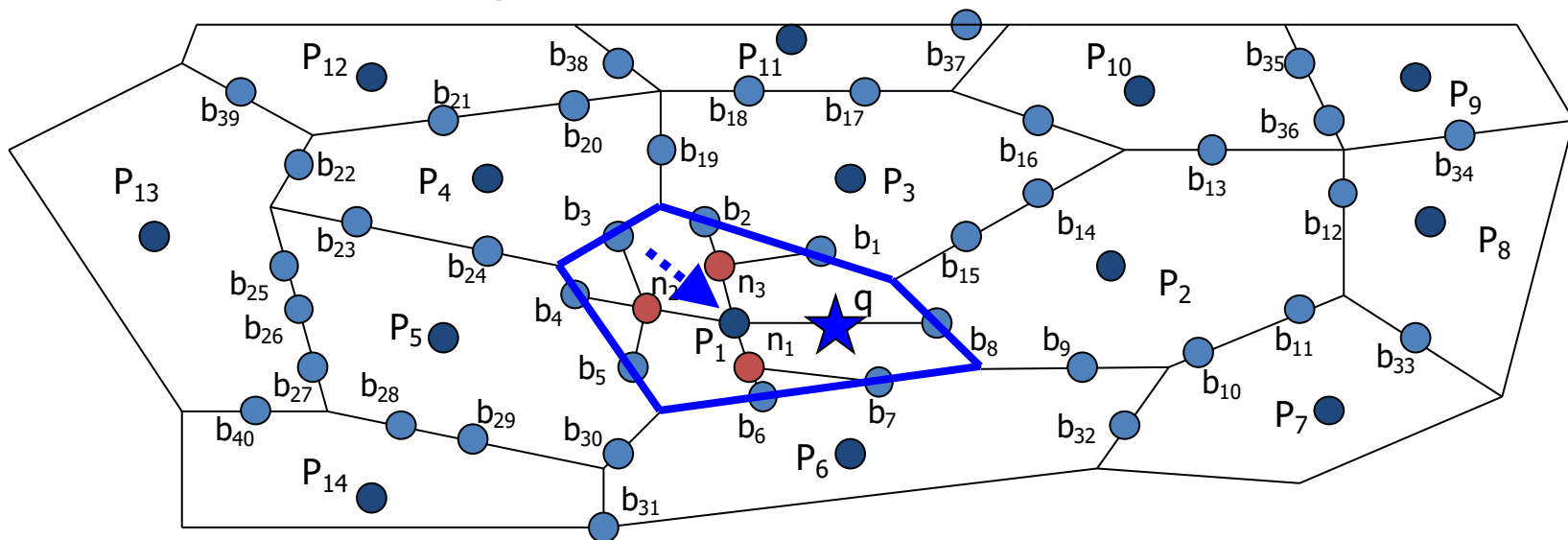
- Find 1st NN
 - Search R-Tree to find the NVP that overlaps q
 - Report the generator of the NVP as the 1st NN;

Cost: $O(\log n)$

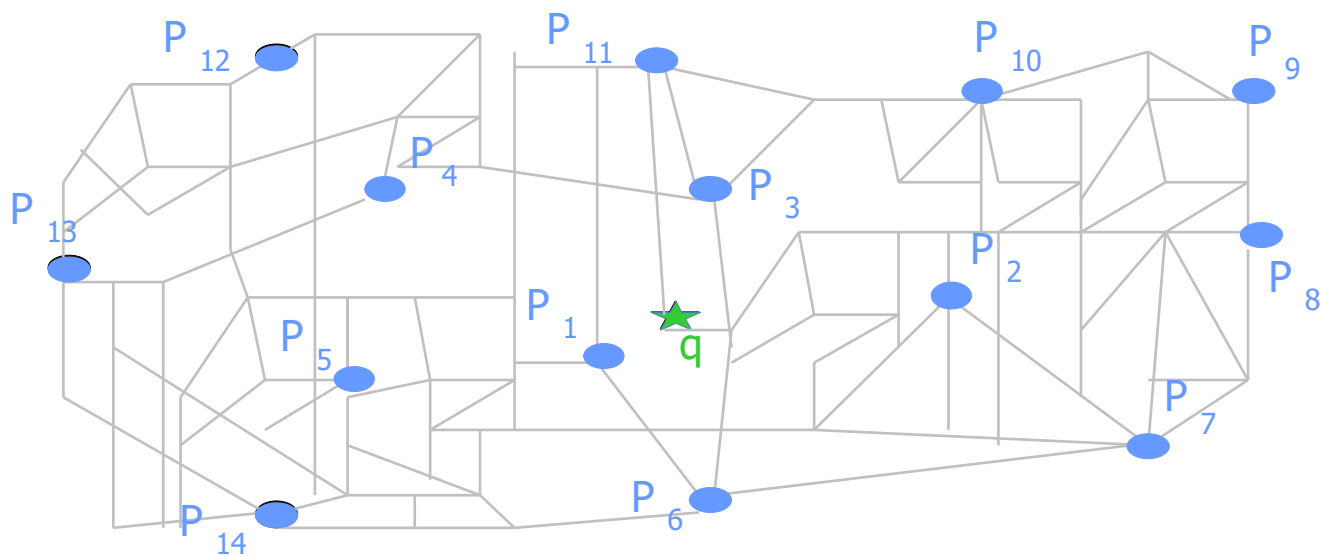


Online Step

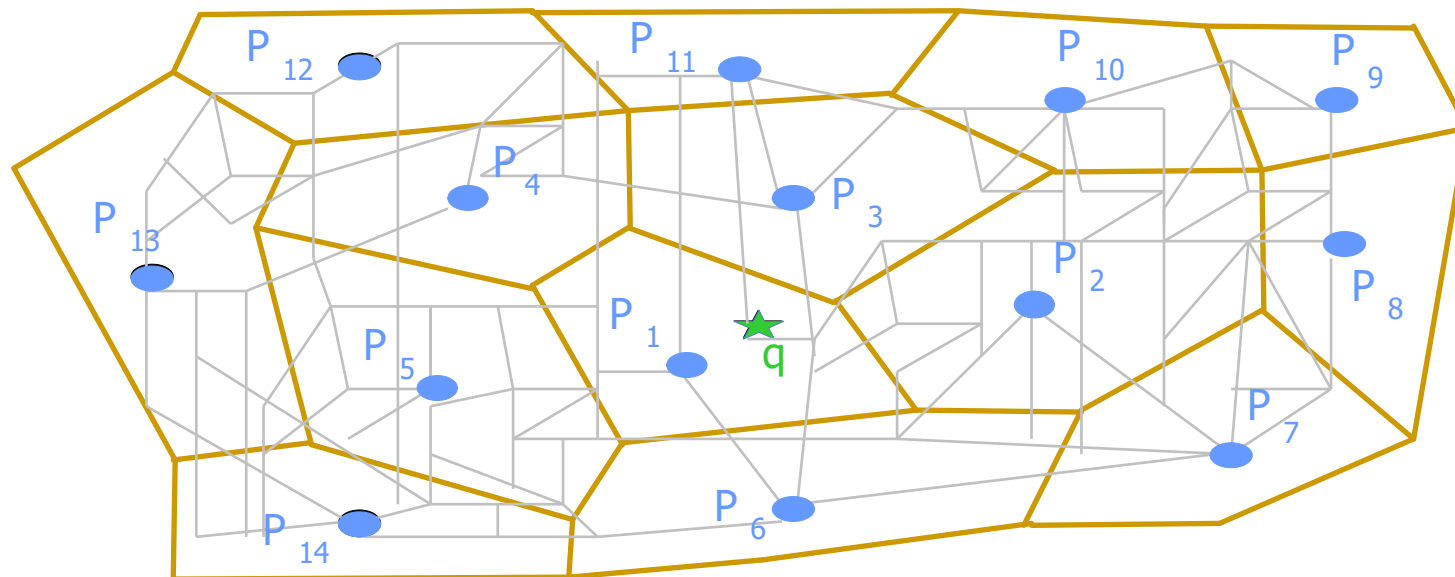
- Find 1st NN
 - Search R-Tree to find the NVP that overlaps q
 - Report the generator of the NVP as the 1st NN;
- Cost: $O(\log n)$



Online Step

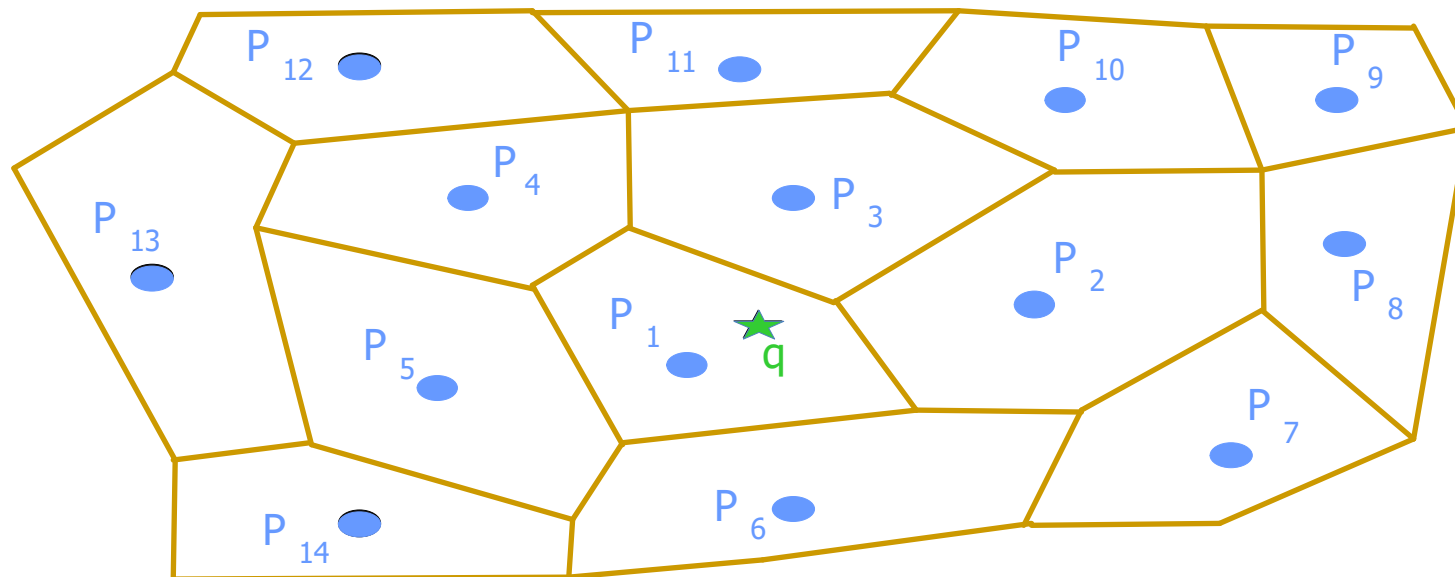


Online Step





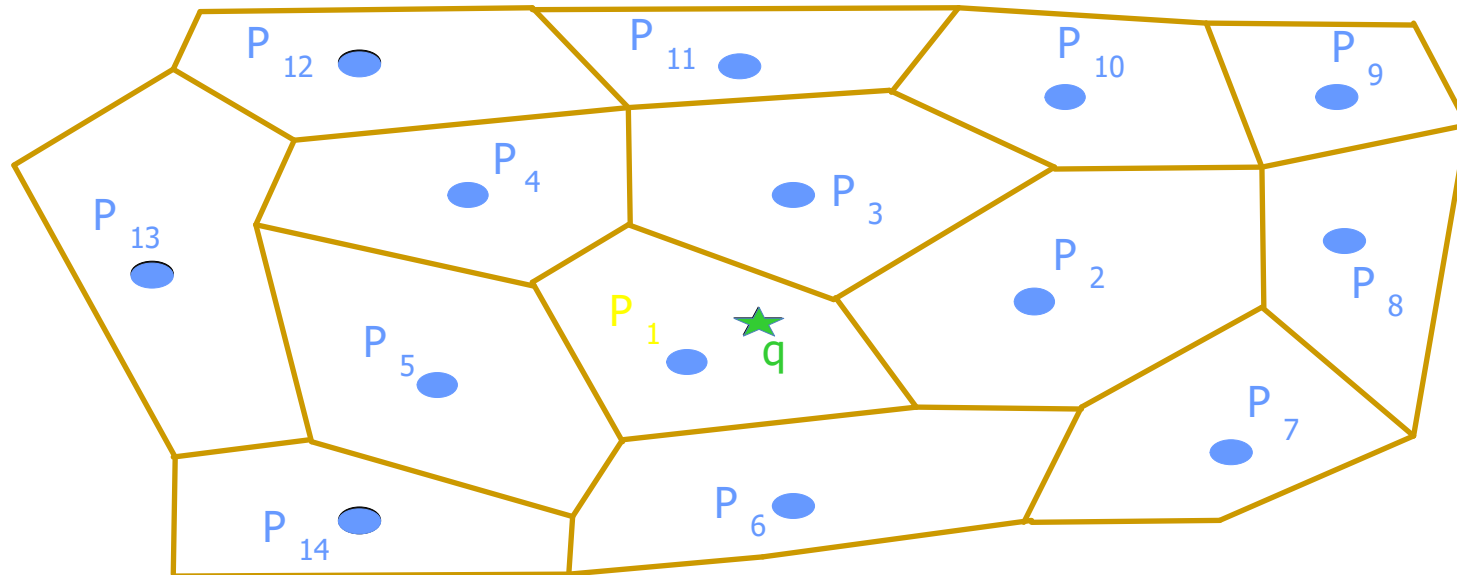
Online Step





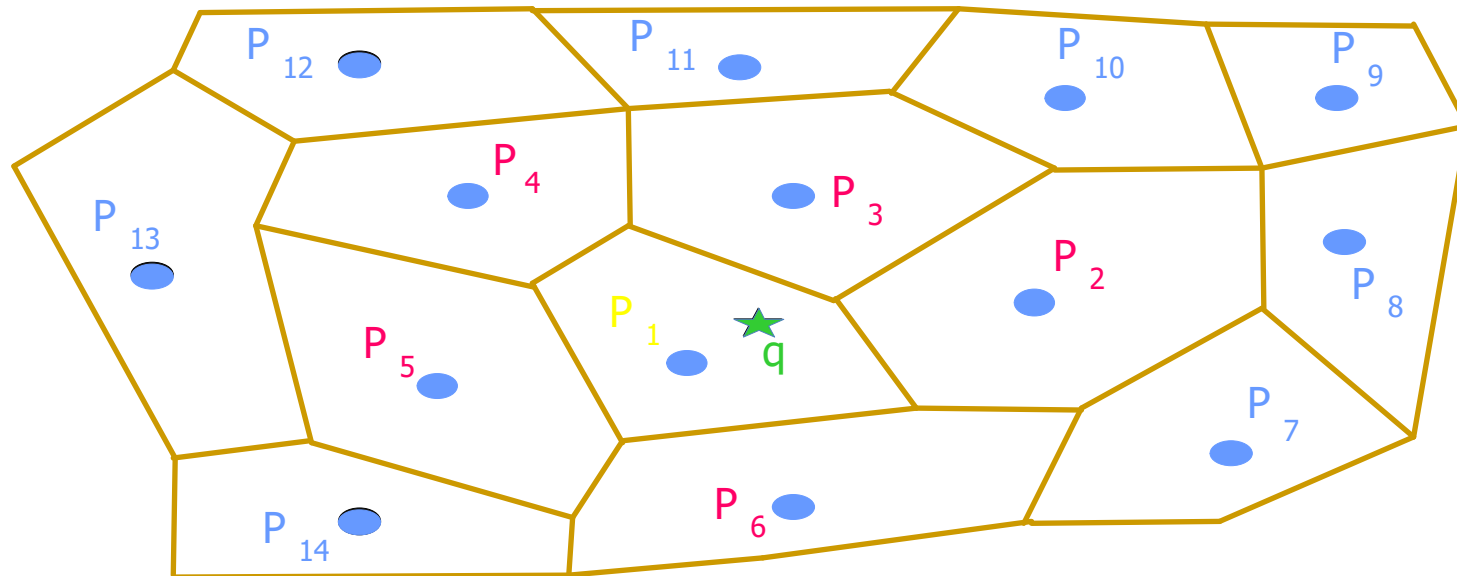
Online Step

- 1st NN = P_1



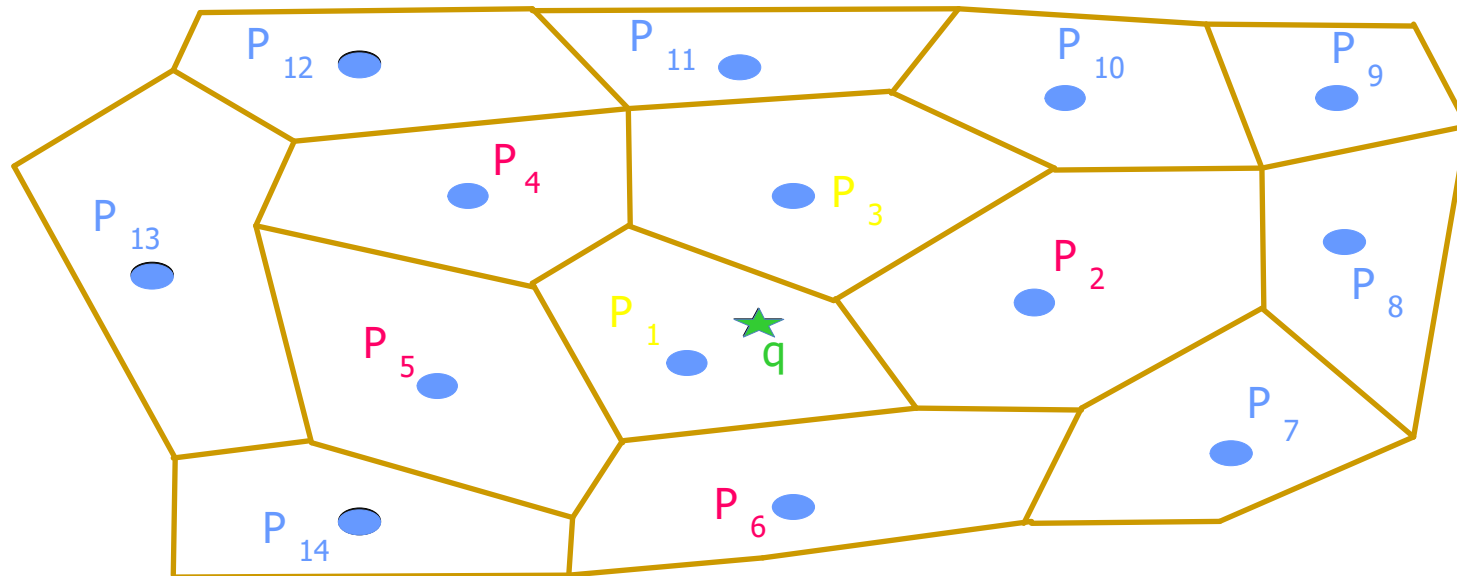
Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$



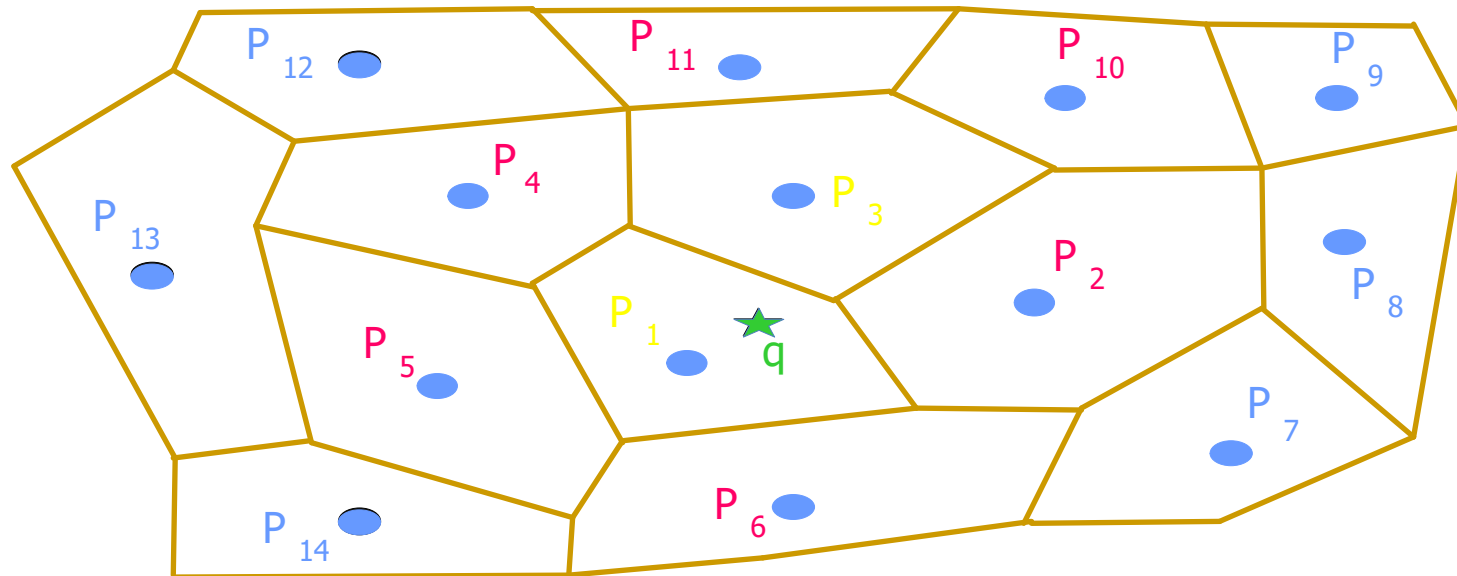
Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)



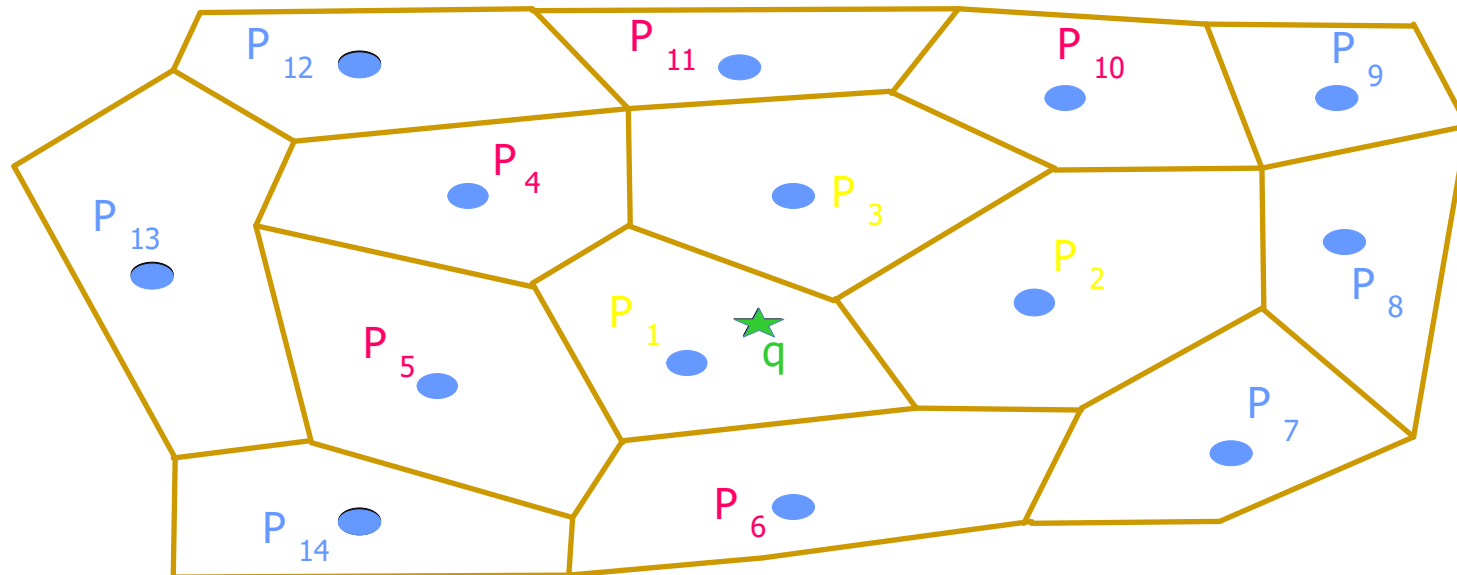
Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_3) \}$



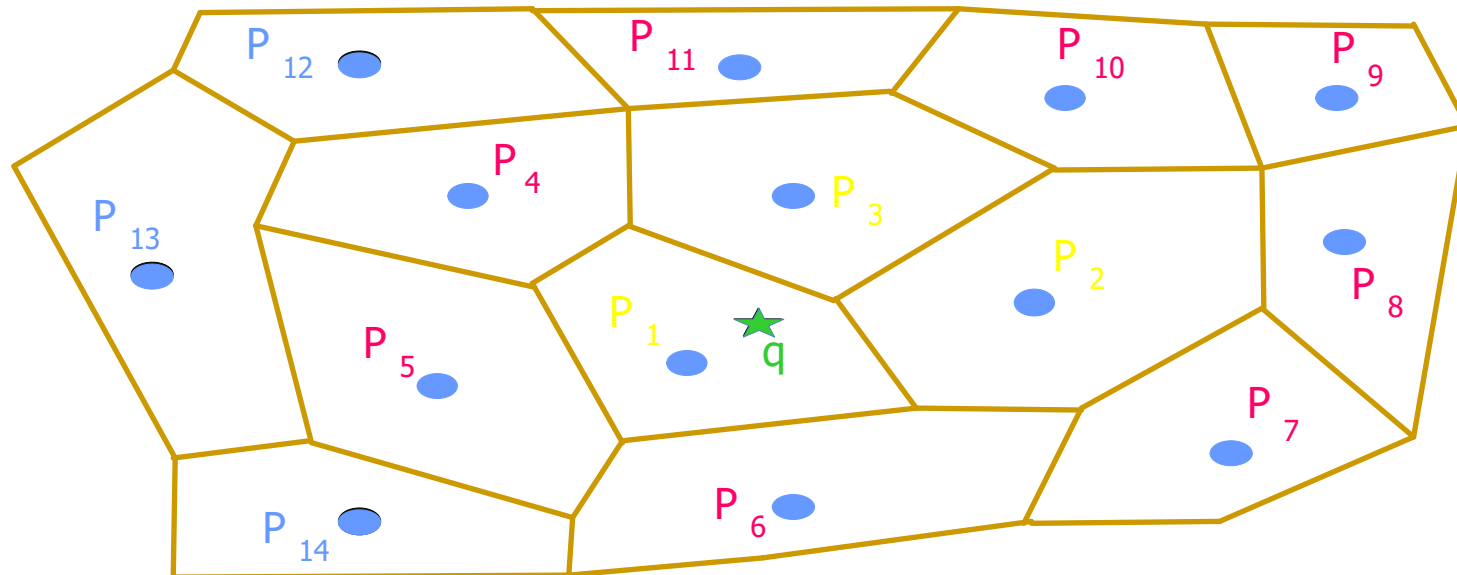
Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_3) \}$
(e.g., P_2)



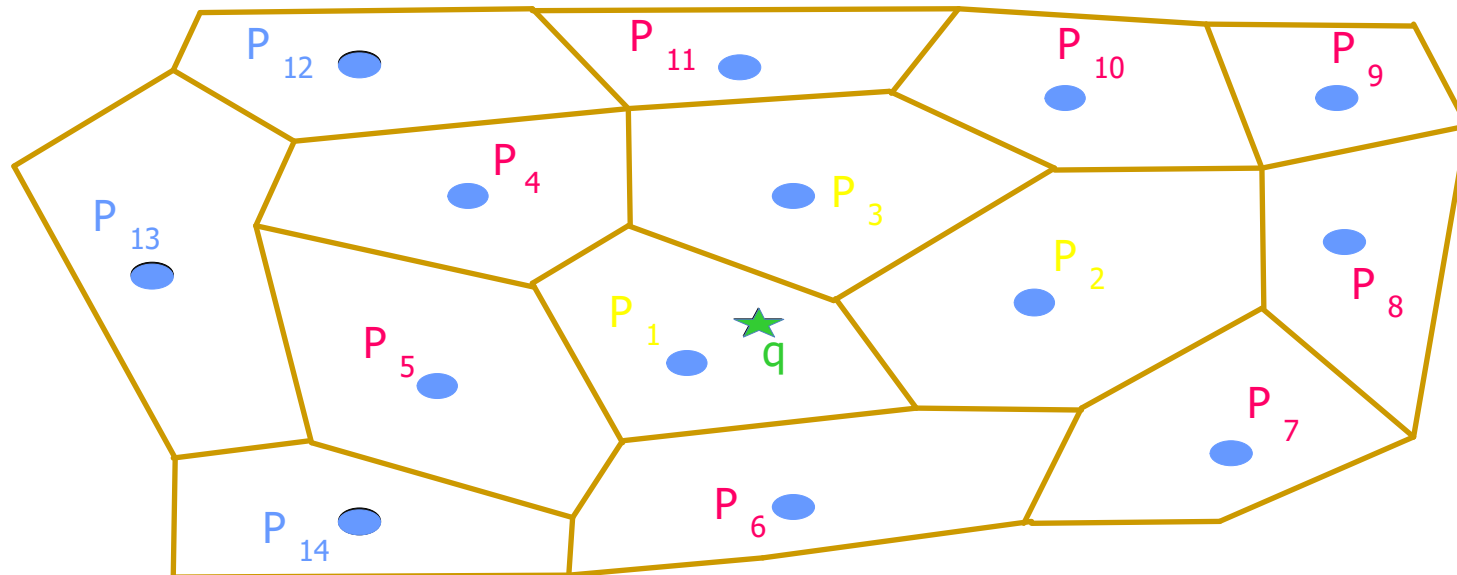
Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_3) \}$
(e.g., P_2)
- 4th NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_3) \cup \text{Neighbors}(P_2) \}$
-



Online Step

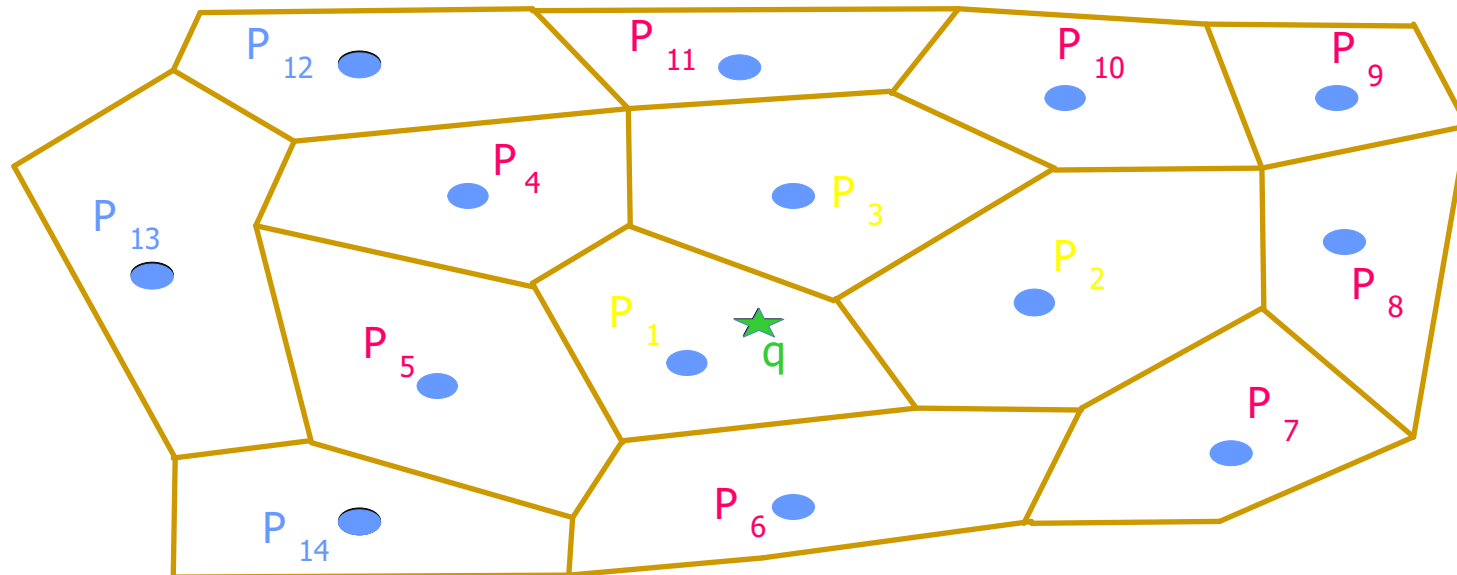
- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
(e.g., P_2)
- 4th NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
-



Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
(e.g., P_2)
- 4th NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
-

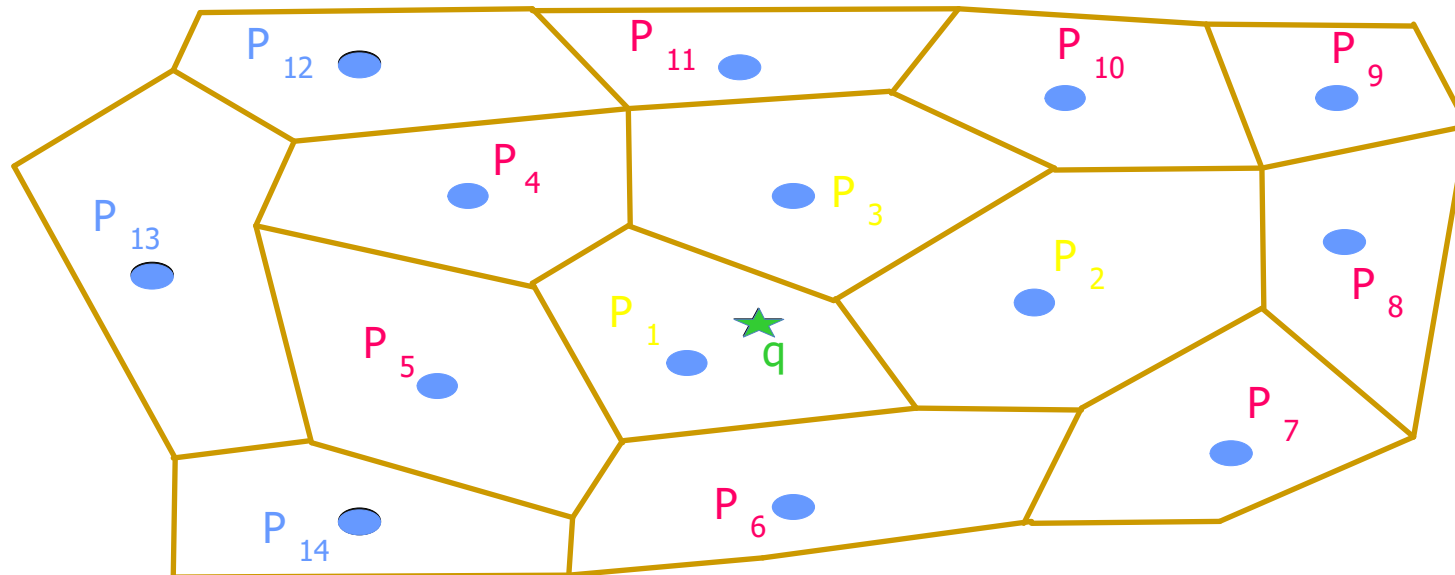
GOOD NEWS! In theory average # of neighbors = $O(5k+1) = O(k)$



Online Step

- 1st NN = P_1
- 2nd NN $\in \{ \text{Neighbors}(P_1) \}$
(e.g., P_3)
- 3rd NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
(e.g., P_2)
- 4th NN $\in \{ \text{Neighbors}(P_1) \cup \text{Neighbors}(P_2) \}$
-

GOOD NEWS! In theory average # of neighbors = $O(5k+1) = O(k)$
In Practice: $\ll 5 \times K$





Performance of VN³

- Data set:
 - Road network in Los Angeles (from NavTeq)
 - 110,000 streets, 79,800 intersections
 - Different points of interest:
 - restaurants, auto services, schools, parks, shopping centers, hospitals
- Measured:
 - Query response time and CPU
 - Comparison with INE [Papadias et al. (vldb03)]
 - Size of candidate set of VN³ filter
 - Comparison with R-tree-based KNN [Seidl et al. (sigmod98) and Hjaltason et al. (tods99)]

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Bigger/less NVPs when sparse 

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Smaller/more NVPs when dense 

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

More Border point 

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Almost same



Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Increases when sparse 

Pre-computation Overhead of VN³

Entities	Cardinality	Points Inside each NVP	Average BPs per NVP	No. of Pre-comp. Border to Border	No. of Pre-comp. Border to inside
Hospitals	0.04%	1698	52	232,000	8,781,000
Shopping	0.16%	458	25	225,600	4,653,000
Parks	0.53%	142	14	239,500	2,630,000
Schools	1.15%	64	10	246,000	1,787,000
Auto. Svc.	3.26%	38	7	239,900	1,611,000
Restaurants	5.80%	27	6	243,600	1,348,000

Naive approach: 3.2 billion pre-computations



Comparison of VN³ and INE

- K = 1

NavTeq Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(0) , 0.018	(0.30) , 12.4
Shopping	0.16% , 173	(0) , 0.020	(0.090) , 3.6
Parks	0.53% , 561	(0) , 0.021	(0.030) , 1.4
Schools	1.15% , 1230	(0) , 0.027	(0.015) , 0.6
Auto. Svc.	3.26% , 2093	(0) , 0.030	(0.013) , 0.67
Restaurants	5.80% , 2944	(0) , 0.032	(0.010) , 0.49



Comparison of VN³ and INE

- K = 1

NavTeq Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(0) , 0.018	(0.30) , 12.4
Shopping	0.16% , 173	(0) , 0.020	(0.090) , 3.6
Parks	0.53% , 561	(0) , 0.021	(0.030) , 1.4
Schools	1.15% , 1230	(0) , 0.027	(0.015) , 0.6
Auto. Svc.	3.26% , 2093	(0) , 0.030	(0.013) , 0.67
Restaurants	5.80% , 2944	(0) , 0.032	(0.010) , 0.49

Almost instantly



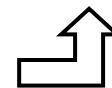


Comparison of VN³ and INE

- K = 1

NavTeq Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(0) , 0.018	(0.30) , 12.4
Shopping	0.16% , 173	(0) , 0.020	(0.090) , 3.6
Parks	0.53% , 561	(0) , 0.021	(0.030) , 1.4
Schools	1.15% , 1230	(0) , 0.027	(0.015) , 0.6
Auto. Svc.	3.26% , 2093	(0) , 0.030	(0.013) , 0.67
Restaurants	5.80% , 2944	(0) , 0.032	(0.010) , 0.49

0 cpu time!





Comparison of VN³ and INE

- K = 1

NavTeq Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(0) , 0.018	(0.30) , 12.4
Shopping	0.16% , 173	(0) , 0.020	(0.090) , 3.6
Parks	0.53% , 561	(0) , 0.021	(0.030) , 1.4
Schools	1.15% , 1230	(0) , 0.027	(0.015) , 0.6
Auto. Svc.	3.26% , 2093	(0) , 0.030	(0.013) , 0.67
Restaurants	5.80% , 2944	(0) , 0.032	(0.010) , 0.49

Increases for sparse



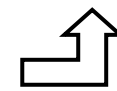


Comparison of VN³ and INE

- K = 5

Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(1.5) , 6.5	(1.7) , 78.3
Shopping	0.16% , 173	(0.45) , 3.3	(0.5) , 21.1
Parks	0.53% , 561	(0.15) , 1.5	(0.2) , 8.2
Schools	1.15% , 1230	(0.06) , 0.75	(0.07) , 3.5
Auto. Svc.	3.26% , 2093	(0.01) , 0.47	(0.05) , 2.43
Restaurants	5.80% , 2944	(0.01) , 0.87	(0.03) , 1.5

VN³ cpu time comparable to INE

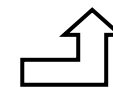
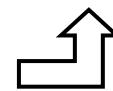


Comparison of VN³ and INE

- K = 5

Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(1.5) , 6.5	(1.7) , 78.3
Shopping	0.16% , 173	(0.45) , 3.3	(0.5) , 21.1
Parks	0.53% , 561	(0.15) , 1.5	(0.2) , 8.2
Schools	1.15% , 1230	(0.06) , 0.75	(0.07) , 3.5
Auto. Svc.	3.26% , 2093	(0.01) , 0.47	(0.05) , 2.43
Restaurants	5.80% , 2944	(0.01) , 0.87	(0.03) , 1.5

Database time still dominant



Comparison of VN³ and INE

- K = 5

Entities	Density, Cardinality	VN ³ (cpu) , db sec	INE (cpu) , disk sec
Hospitals	0.04% , 46	(1.5) , 6.5	(1.7) , 78.3
Shopping	0.16% , 173	(0.45) , 3.3	(0.5) , 21.1
Parks	0.53% , 561	(0.15) , 1.5	(0.2) , 8.2
Schools	1.15% , 1230	(0.06) , 0.75	(0.07) , 3.5
Auto. Svc.	3.26% , 2093	(0.01) , 0.47	(0.05) , 2.43
Restaurants	5.80% , 2944	(0.01) , 0.87	(0.03) , 1.5

👍 Excluding K=1, VN3 is between 2 (high densities / larger K) to 14 (low densities / larger K) times faster than INE

VN³ vs INE

		Query Processing Time (Sec)											
		K=1		K=5		K=10		K=25		K=50		K=100	
		VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE
Entities	Qty (density)	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk
Hospital	46 (0.0004)	(0) 0.018	(0.3) 12.4	(1.5) 6.5	(1.7) 78.3	(4.5) 14.0	(3.8) 165.1	(15.3) 35.1	(10.1) 430.2	- -	- -	- -	- -
Shopping Centers	173 (0.0016)	(0) 0.020	(0.09) 3.6	(0.45) 3.3	(0.5) 21.1	(1.3) 6.9	(1.1) 44.0	(3.4) 18.1	(3.1) 118.0	- -	- -	- -	- -
Parks	561 (0.0053)	(0) 0.021	(0.03) 1.4	(0.15) 1.5	(0.2) 8.2	(0.37) 2.8	(0.3) 15.3	(1.4) 6.4	(0.8) 36.4	(2.5) 13.3	(1.6) 71.1	- -	- -
Schools	1230 (0.0115)	(0) 0.027	(0.015) 0.6	(0.06) 0.75	(0.07) 3.5	(0.18) 1.46	(0.14) 6.6	(0.7) 3.9	(0.36) 15.6	(1.9) 7.5	(0.7) 32.2	- -	- -
Auto Services	2093 (0.0326)	(0) 0.030	(0.013) 0.57	(0.01) 0.65	(0.05) 2.43	(0.09) 1.4	(0.09) 4.3	(0.58) 2.95	(0.23) 10.0	(1.65) 6.68	(0.44) 19.4	(2.78) 13.1	(0.87) 38.00
Restaurants	2944 (0.0580)	(0) 0.032	(0.01) 0.49	(0.01) 0.57	(0.03) 1.34	(0.04) 1.48	(0.06) 2.7	(0.26) 2.8	(0.15) 6.8	(0.8) 6.1	(0.3) 13.3	(1.85) 12.8	(0.6) 26.0

VN³ vs INE

VN³ finds the 1st NN using R-Tree
INE needs to expand the network around q

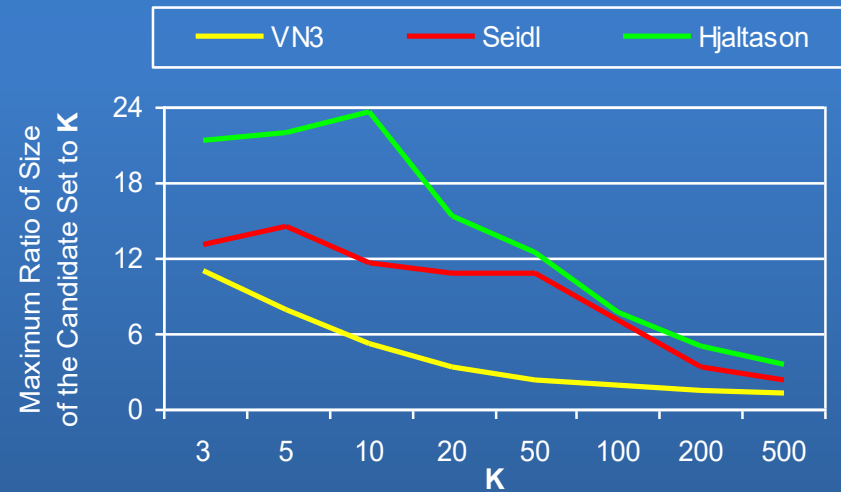
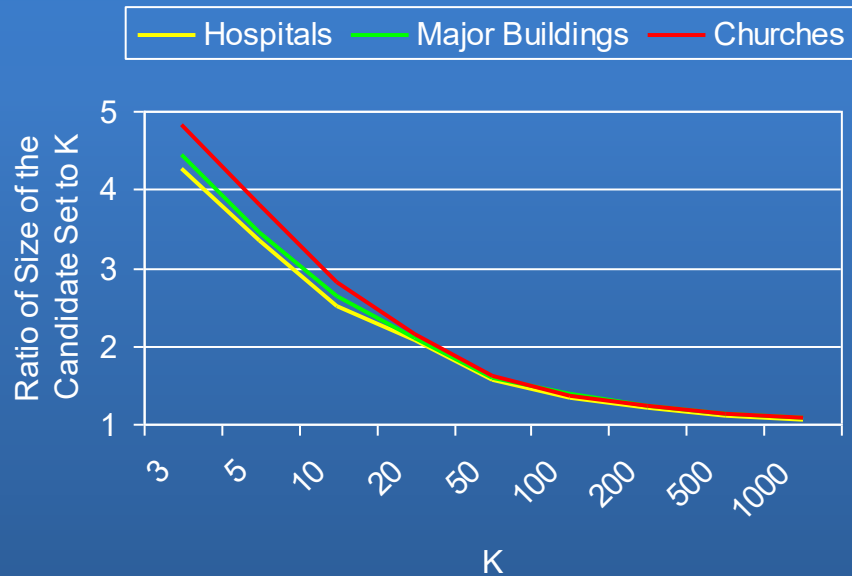
		Query Processing Time (Sec)											
		K=1		K=5		K=10		K=25		K=50		K=100	
Entities	Qty (density)	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE
		(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk
Hospital	46 (0.0004)	(0) 0.018	(0.3) 12.4	(1.5) 6.5	(1.7) 78.3	(4.5) 14.0	(3.8) 165.1	(15.3) 35.1	(10.1) 430.2	- -	- -	- -	- -
Shopping Centers	173 (0.0016)	(0) 0.020	(0.09) 3.6	(0.45) 3.3	(0.5) 21.1	(1.3) 6.9	(1.1) 44.0	(3.4) 18.1	(3.1) 118.0	- -	- -	- -	- -
Parks	561 (0.0053)	(0) 0.021	(0.03) 1.4	(0.15) 1.5	(0.2) 8.2	(0.37) 2.8	(0.3) 15.3	(1.4) 6.4	(0.8) 36.4	(2.5) 13.3	(1.6) 71.1	- -	- -
Schools	1230 (0.0115)	(0) 0.027	(0.015) 0.6	(0.06) 0.75	(0.07) 3.5	(0.18) 1.46	(0.14) 6.6	(0.7) 3.9	(0.36) 15.6	(1.9) 7.5	(0.7) 32.2	- -	- -
Auto Services	2093 (0.0326)	(0) 0.030	(0.013) 0.57	(0.01) 0.65	(0.05) 2.43	(0.09) 1.4	(0.09) 4.3	(0.58) 2.95	(0.23) 10.0	(1.65) 6.68	(0.44) 19.4	(2.78) 13.1	(0.87) 38.00
Restaurants	2944 (0.0580)	(0) 0.032	(0.01) 0.49	(0.01) 0.57	(0.03) 1.34	(0.04) 1.48	(0.06) 2.7	(0.26) 2.8	(0.15) 6.8	(0.8) 6.1	(0.3) 13.3	(1.85) 12.8	(0.6) 26.0

VN³ vs INE

CPU Time:
INE uses a queue
which is incrementally
updated

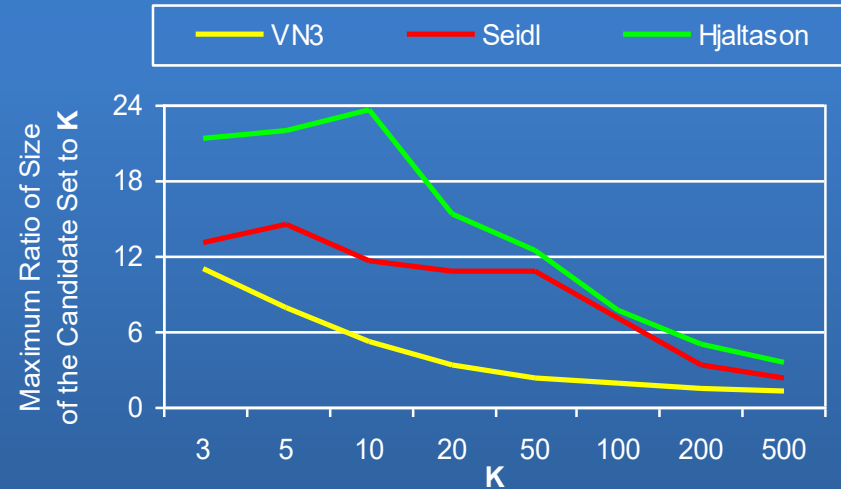
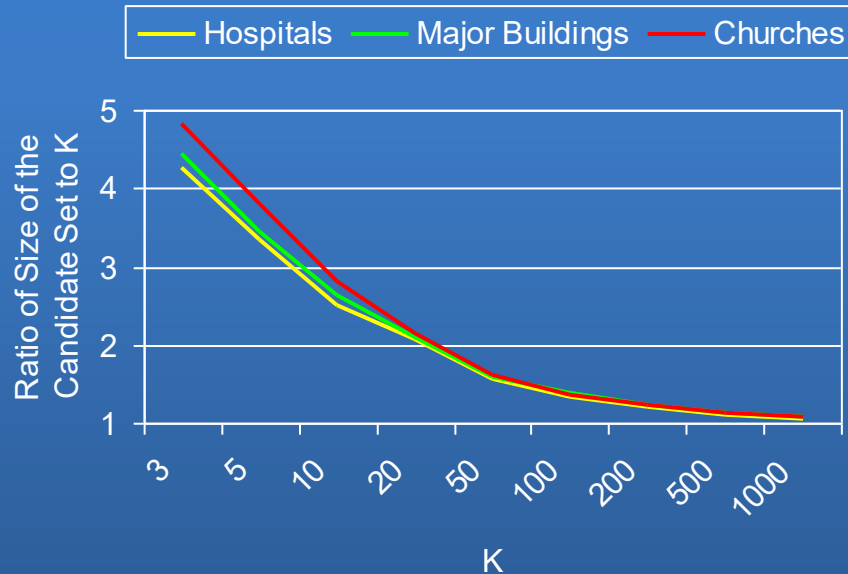
		Query Processing Time (Sec)											
		K=1		K=5		K=10		K=25		K=50		K=100	
		VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE	VN ³	INE
Entities	Qty (density)	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk	(cpu) disk
Hospital	46 (0.0004)	(0) 0.018	(0.3) 12.4	(1.5) 6.5	(1.7) 78.3	(4.5) 14.0	(3.8) 165.1	(15.3) 35.1	(10.1) 430.2	-	-	-	-
Shopping Centers	173 (0.0016)	(0) 0.020	(0.09) 3.6	(0.45) 3.3	(0.5) 21.1	(1.3) 6.9	(1.1) 44.0	(3.4) 18.1	(3.1) 118.0	-	-	-	-
Parks	561 (0.0053)	(0) 0.021	(0.03) 1.4	(0.15) 1.5	(0.2) 8.2	(0.37) 2.8	(0.3) 15.3	(1.4) 6.4	(0.8) 36.4	(2.5) 13.3	(1.6) 71.1	-	-
Schools	1230 (0.0115)	(0) 0.027	(0.015) 0.6	(0.06) 0.75	(0.07) 3.5	(0.18) 1.46	(0.14) 6.6	(0.7) 3.9	(0.36) 15.6	(1.9) 7.5	(0.7) 32.2	-	-
Auto Services	2093 (0.0326)	(0) 0.030	(0.013) 0.57	(0.01) 0.65	(0.05) 2.43	(0.09) 1.4	(0.09) 4.3	(0.58) 2.95	(0.23) 10.0	(1.65) 6.68	(0.44) 19.4	(2.78) 13.1	(0.87) 38.00
Restaurants	2944 (0.0580)	(0) 0.032	(0.01) 0.49	(0.01) 0.57	(0.03) 1.34	(0.04) 1.48	(0.06) 2.7	(0.26) 2.8	(0.15) 6.8	(0.8) 6.1	(0.3) 13.3	(1.85) 12.8	(0.6) 26.0

VN³: Performance of Filter Step



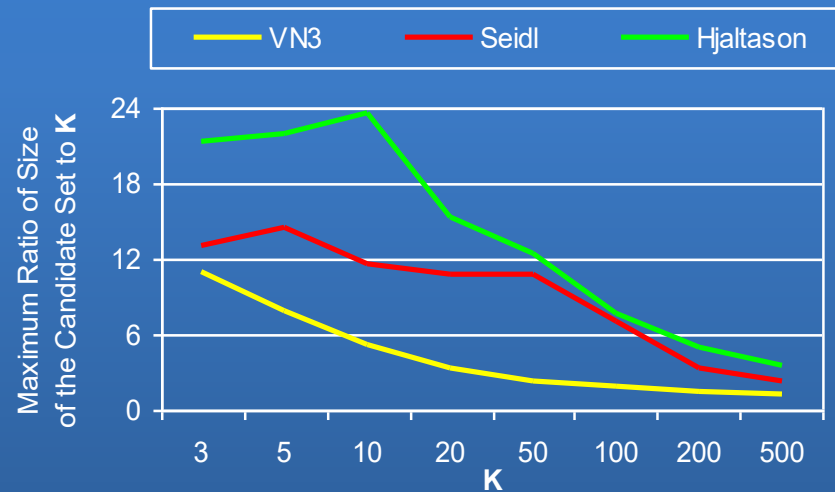
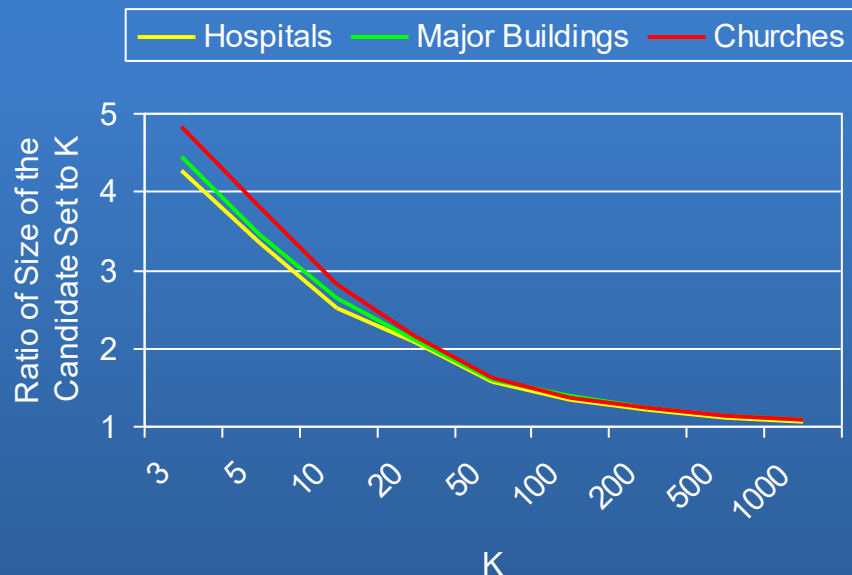
- USGS Data, Euclidean space

VN³: Performance of Filter Step



- USGS Data, Euclidean space
- 👍 Ratio of size of candidate set to K decreases as K increases
 - Some of the new neighbors are already explored!
- 👍 VN³'s filter Behaves independently from density/distribution of points of interest
 - Size of the candidate set ($O(k)$) is independent of the density/distribution

VN³: Performance of Filter Step



- USGS Data, Euclidean space
- 👍 Ratio of size of candidate set to K decreases as K increases
 - Some of the new neighbors are already explored!
- 👍 VN³'s filter Behaves independently from density/distribution of points of interest
 - Size of the candidate set ($O(k)$) is independent of the density/distribution
- 👍 Better selectivity as compared to other approaches (up to 4 times)
- 👍 Very low variance (predictable behavior)

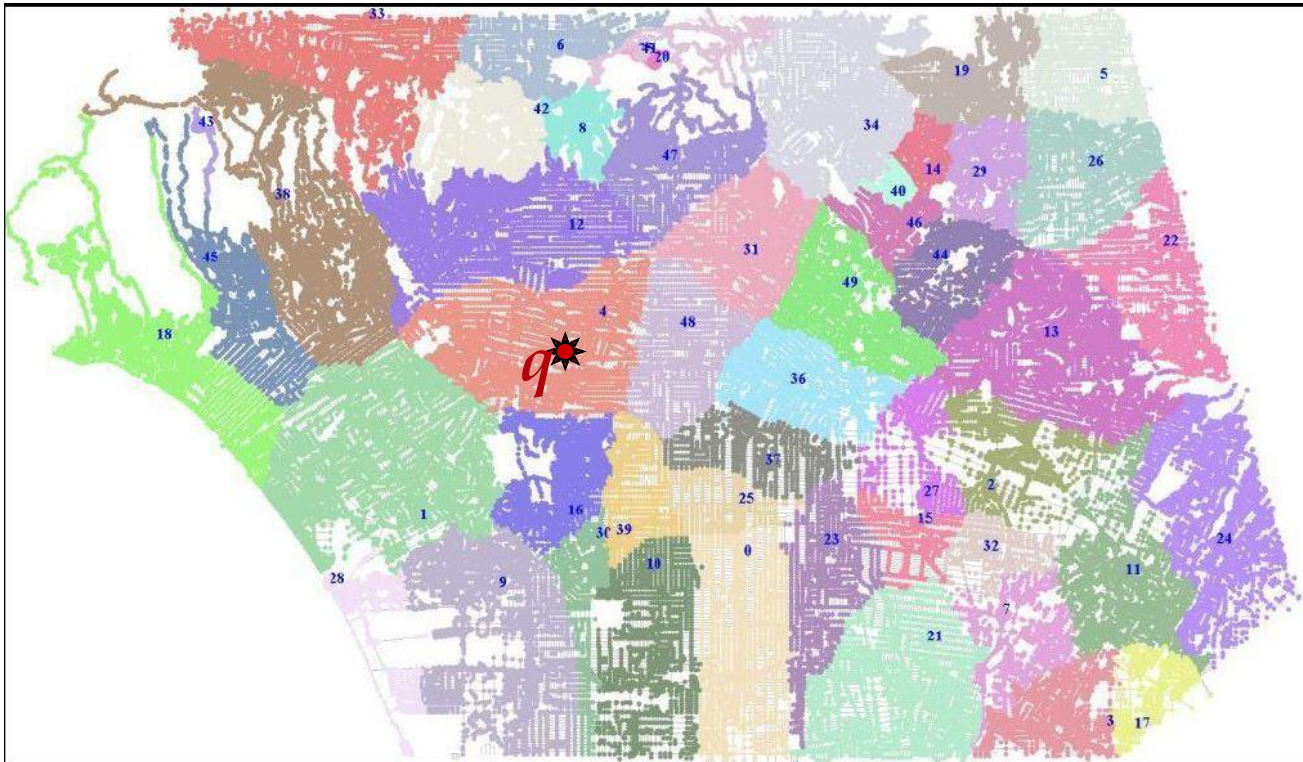


Conclusion

- A novel approach for KNN queries in SNDB
- Based on:
 - Pre-calculating the solution space (first order Voronoi diagrams)
 - Pre-computing some distances (from borders to points inside each polygon)
- Outperforms the only other solution for SNDB
- Independent from object distribution
- Outperforms the solutions for Euclidean space in “filtering”
- Although VN3 is built on complex properties to prove its correctness, but it is a straightforward approach to implement by utilizing simple data structures such as R-tree and lookup tables.

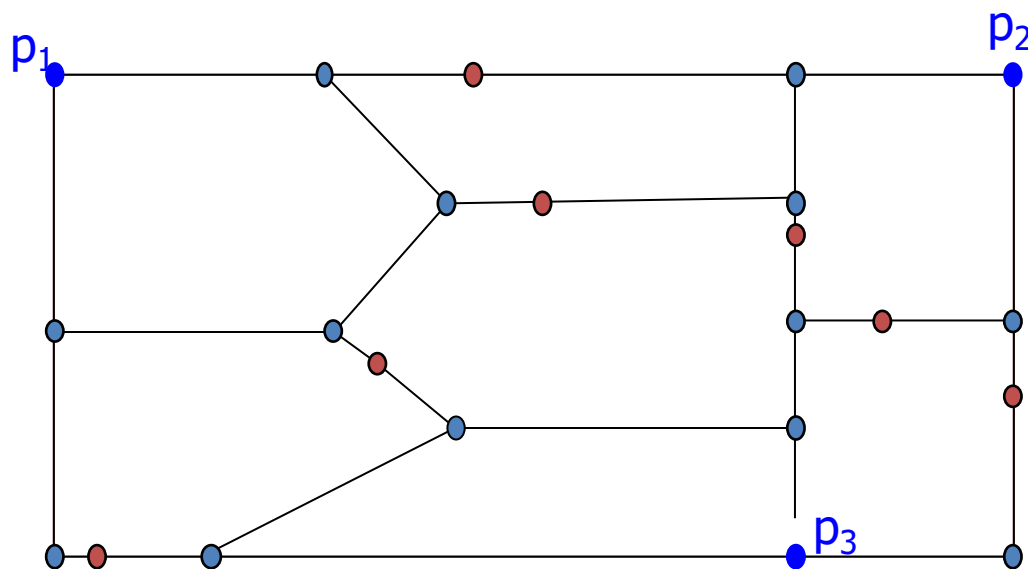
Problem Definition

- Given a NVD and query location q , how can we efficiently find the NVP that overlaps with q ; **contain(q)**?



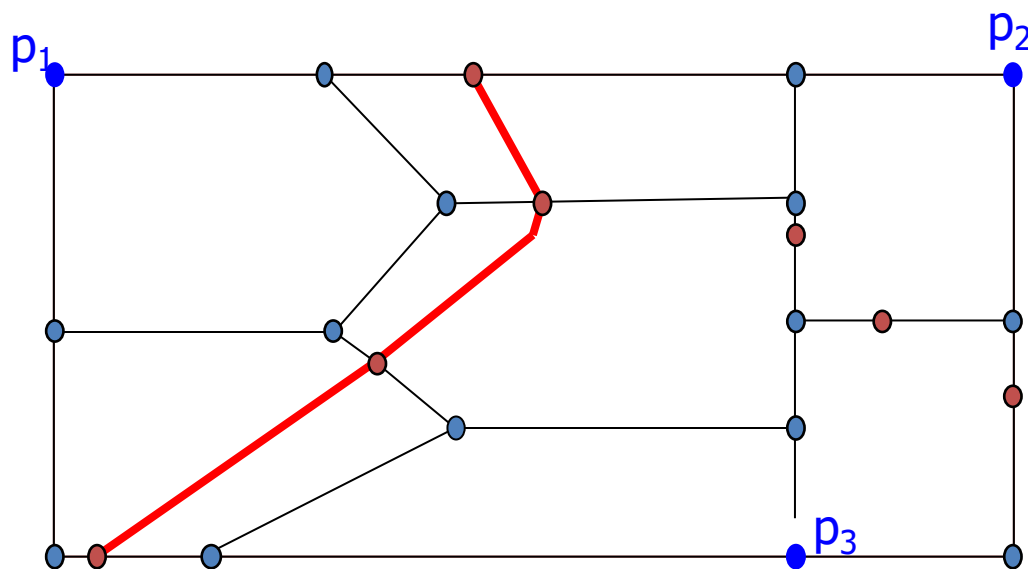
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



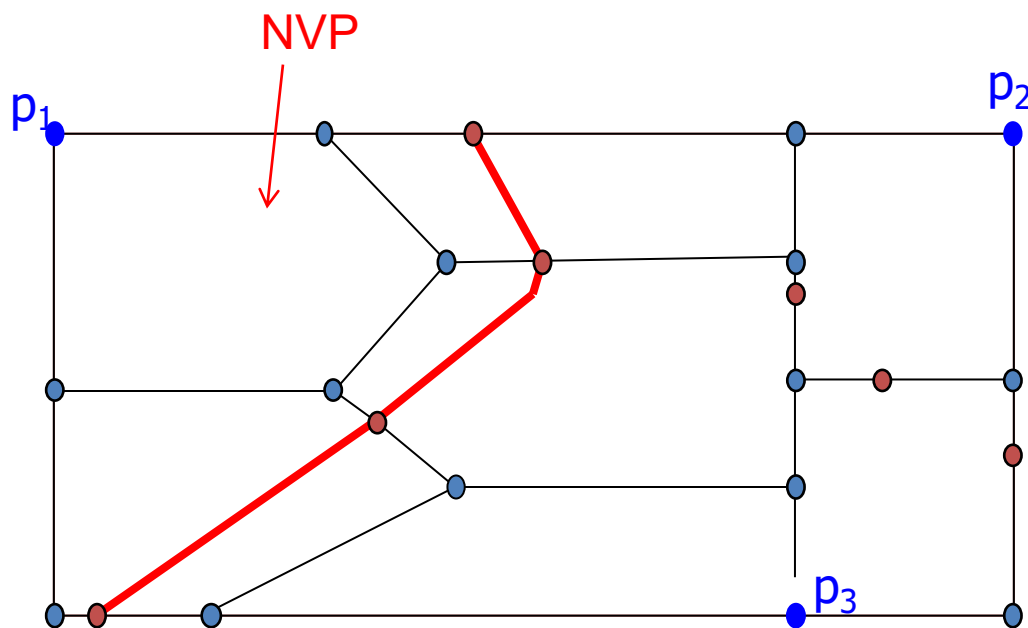
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



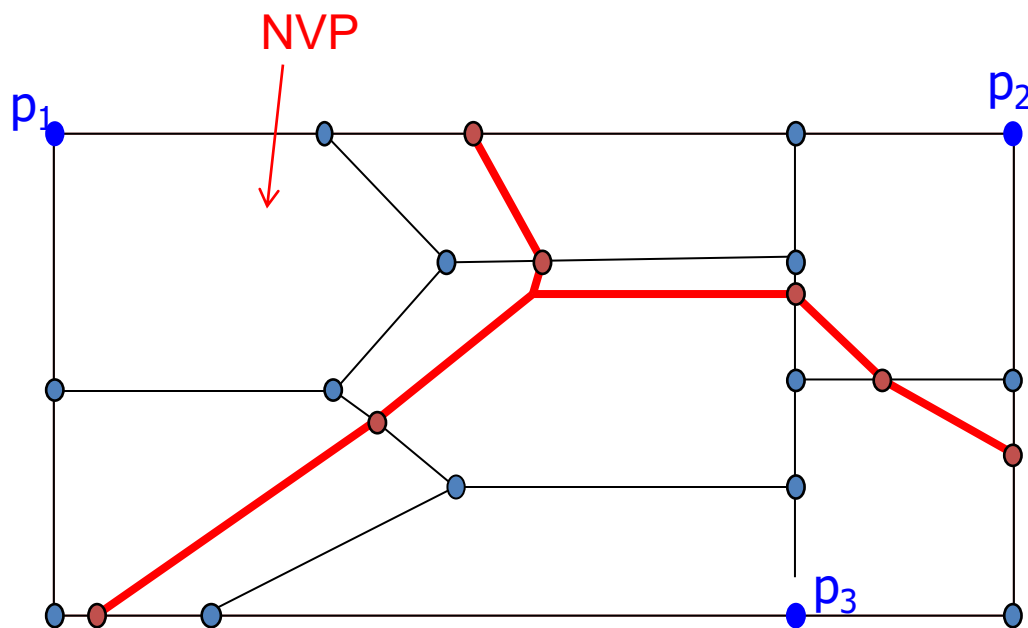
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



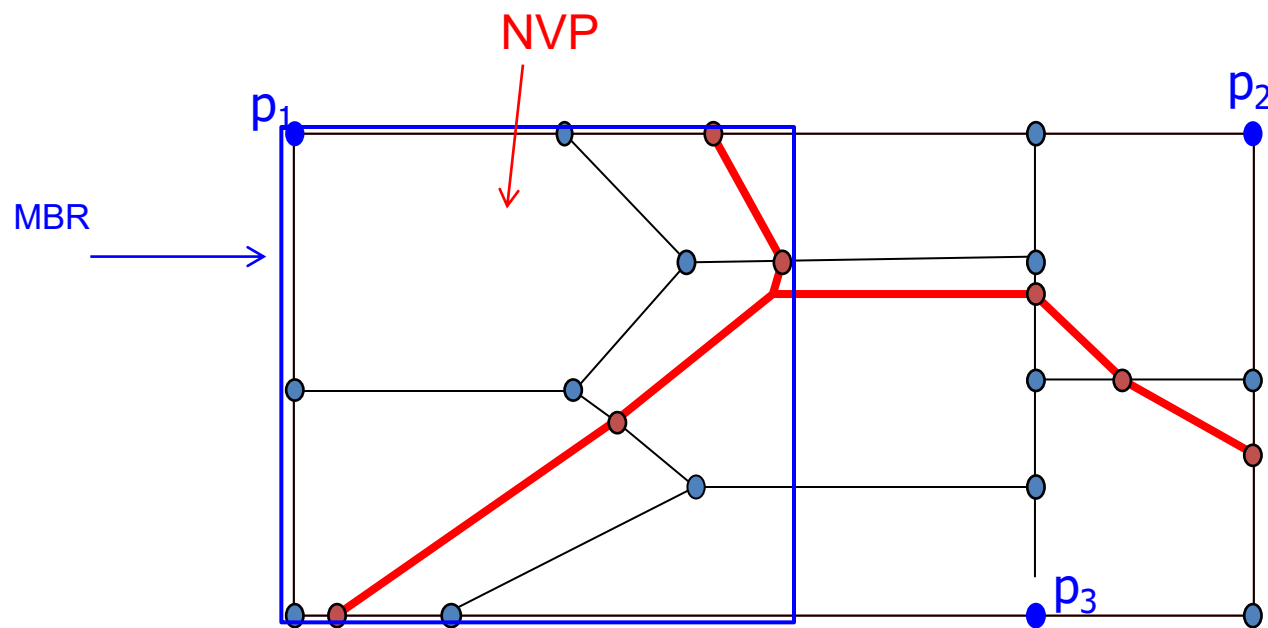
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



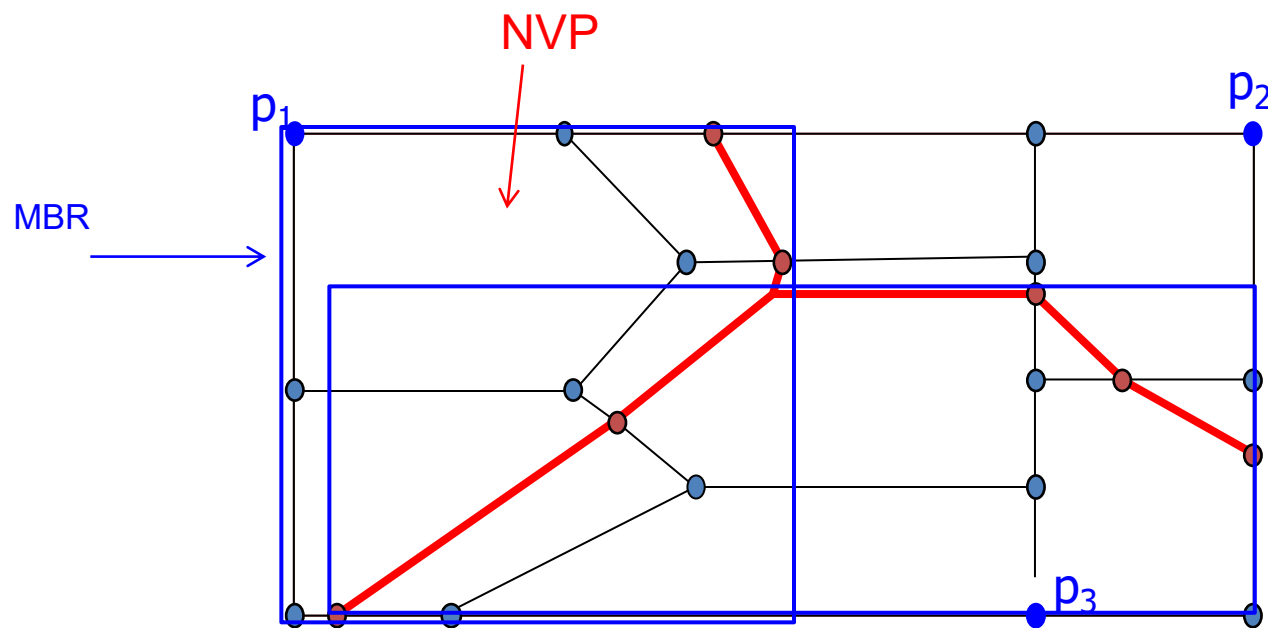
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



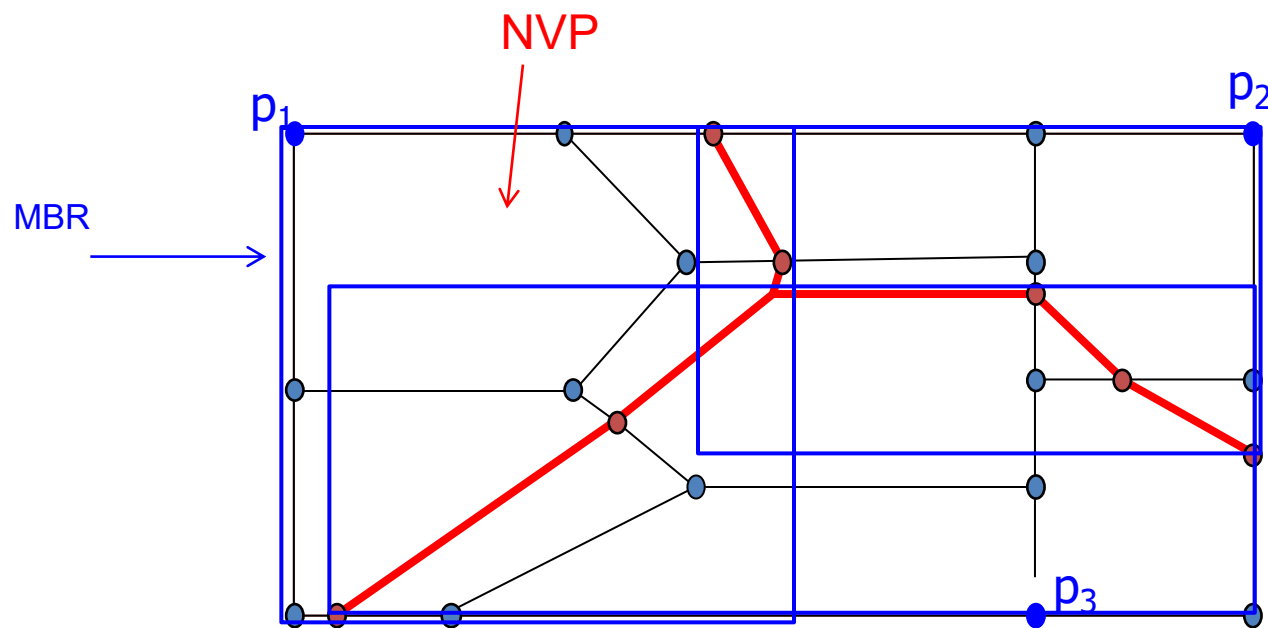
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



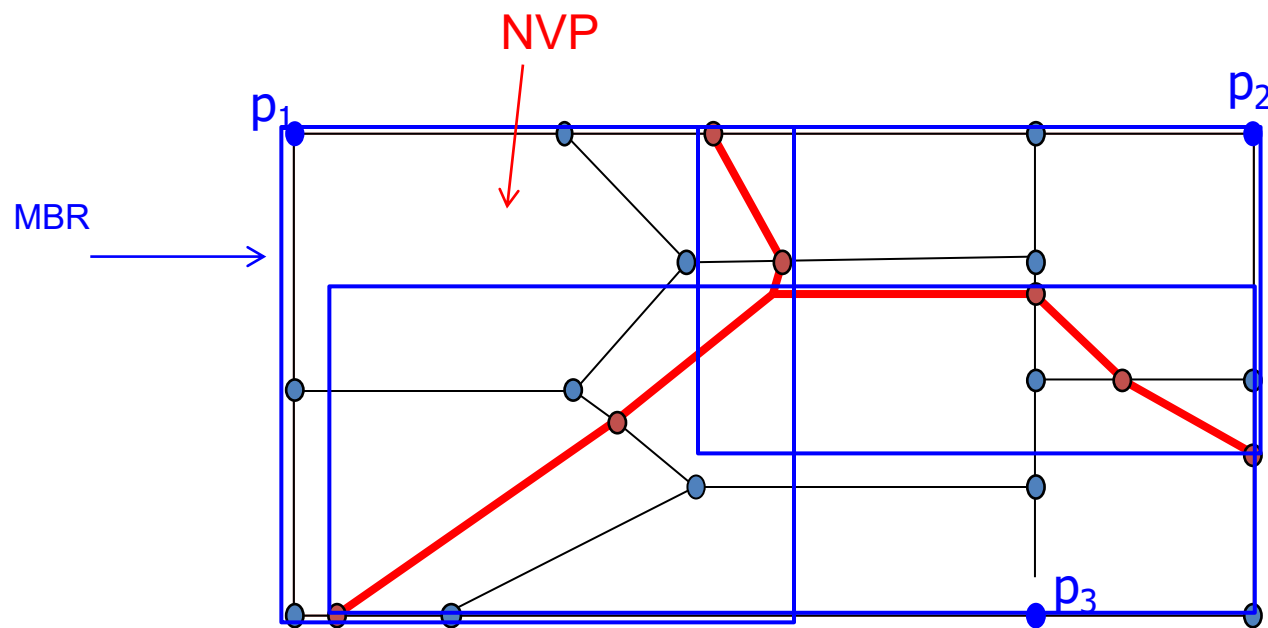
Indexing NVD

- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



Indexing NVD

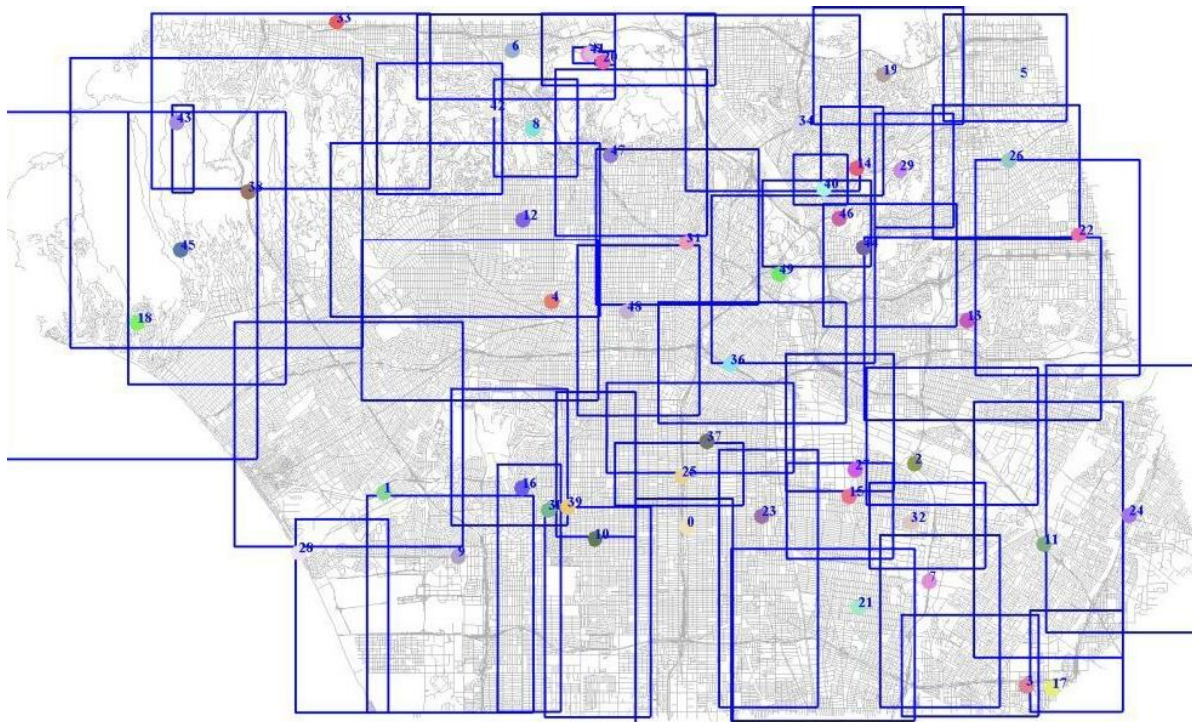
- Voronoi R-tree (VR-tree)
 - Generate NVPs by connecting border points (See Section 4.3)
 - Create minimum bounding rectangle (MBR) around NVPs and index MBRs with R-Tree



VR-tree [Shahabi'04, Taniar'07, Safar'06, Xuan'10]

Indexing NVD

- VR-tree – Problem 1
 - VR-tree splits the network space with hierarchically nested and **largely over-lapping** MBRs
 - **Non-disjoint** partitioning of the space



Indexing NVD

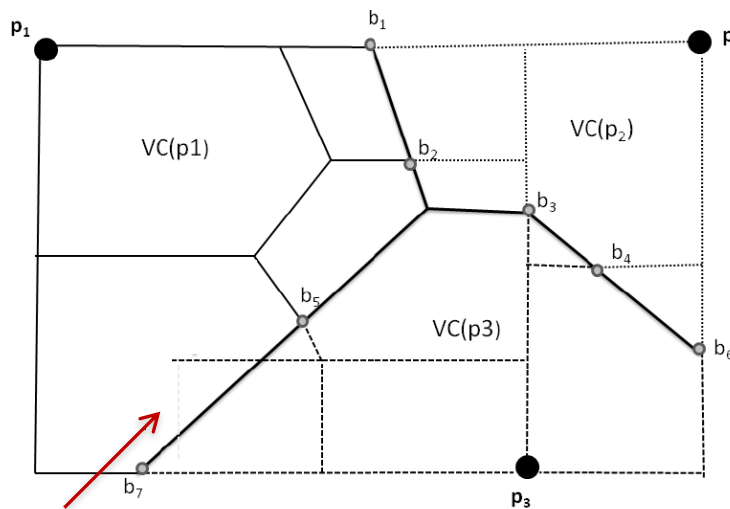
- VR-tree – Problem 1
 - VR-tree splits the network space with hierarchically nested and **largely over-lapping** MBRs
 - **Non-disjoint** partitioning of the space

Indexing NVD

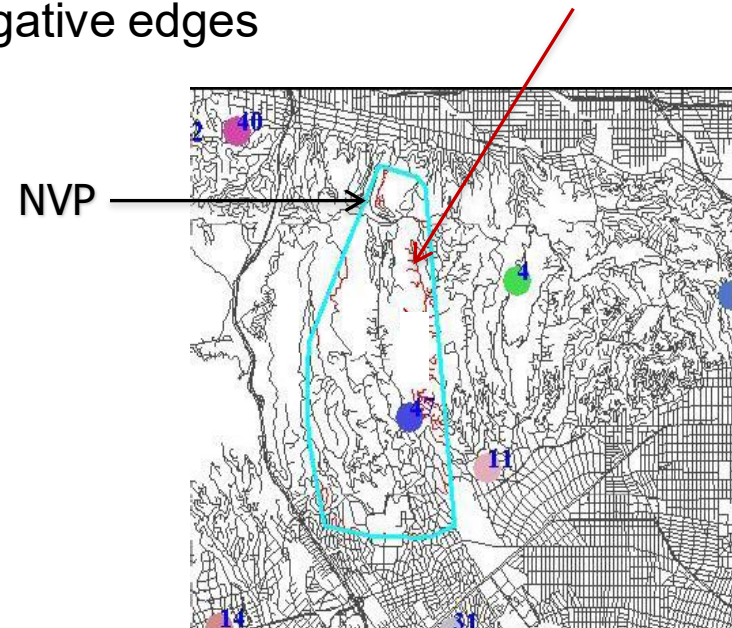
- VR-tree – Problem 2

- Although the NVPs are computed based on the network distance metric, they are treated as polygons in Euclidean space
- Misclassification of edges: False-negative edges

False-negative edges

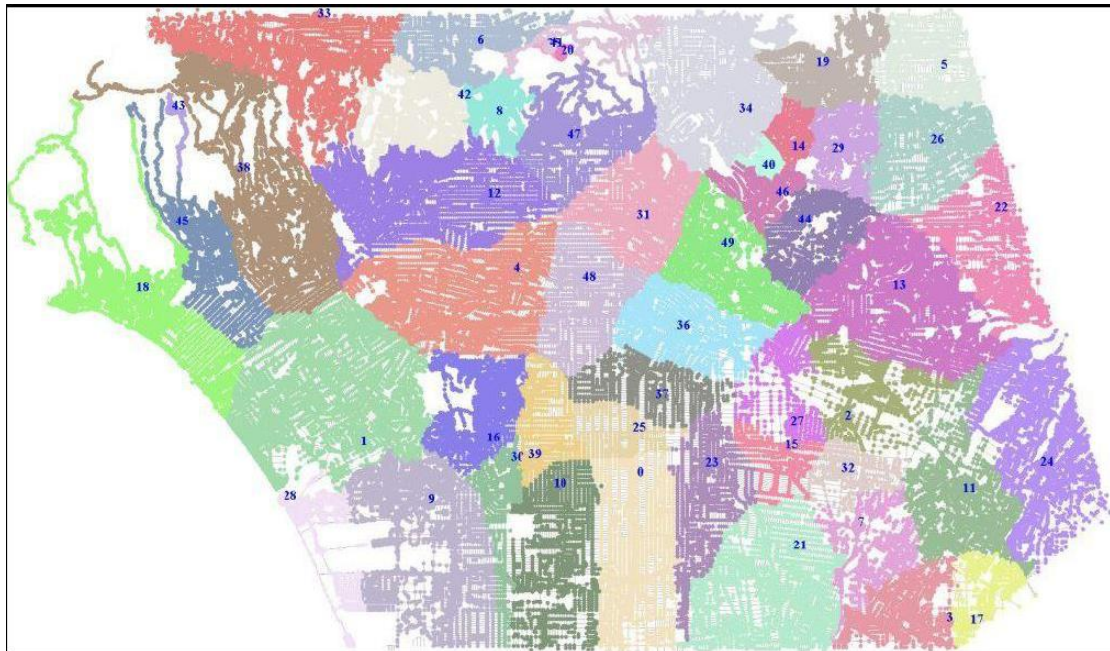


False-negative edges



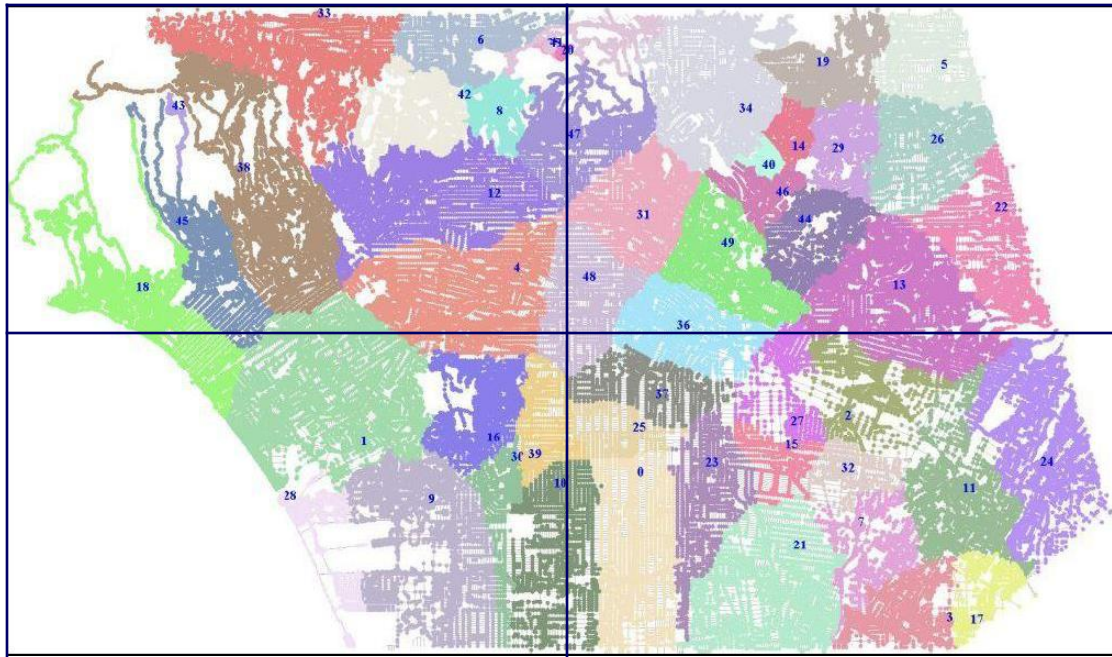
Indexing NVD

- VQ-tree
 - NVPs are
 - **mutually exclusive** : they do not have any overlapping areas
 - **collectively exhaustive**: every node/edges in the network is associated with at least one generator



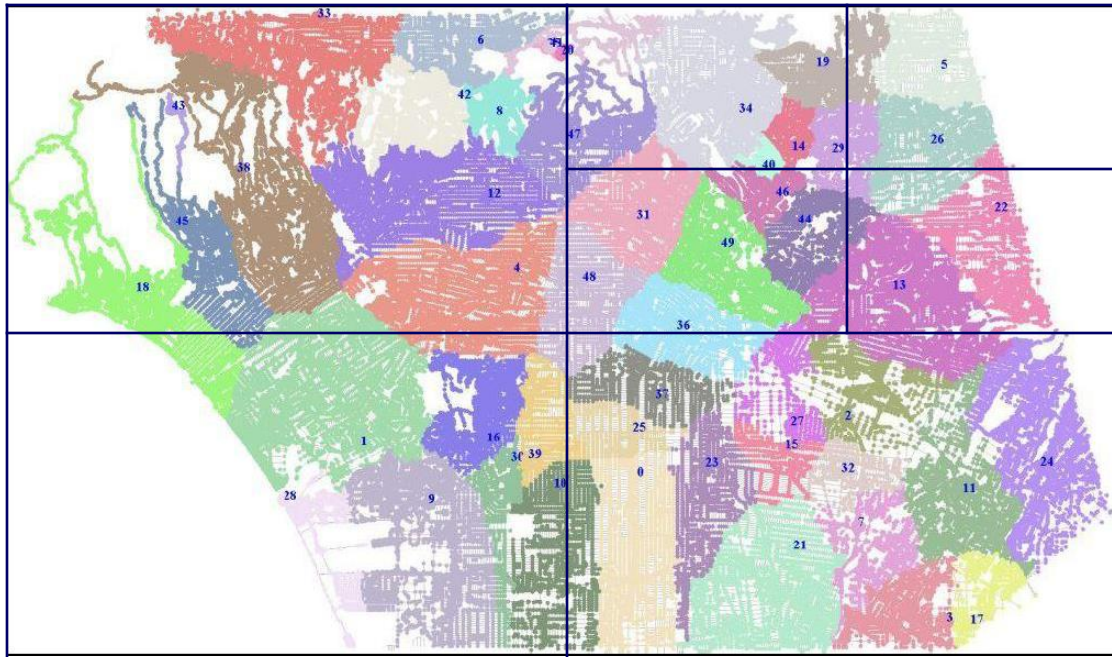
Indexing NVD

- VQ-tree
 - NVPs are
 - **mutually exclusive** : they do not have any overlapping areas
 - **collectively exhaustive**: every node/edges in the network is associated with at least one generator
 - Disjoint decomposition of the NVPs



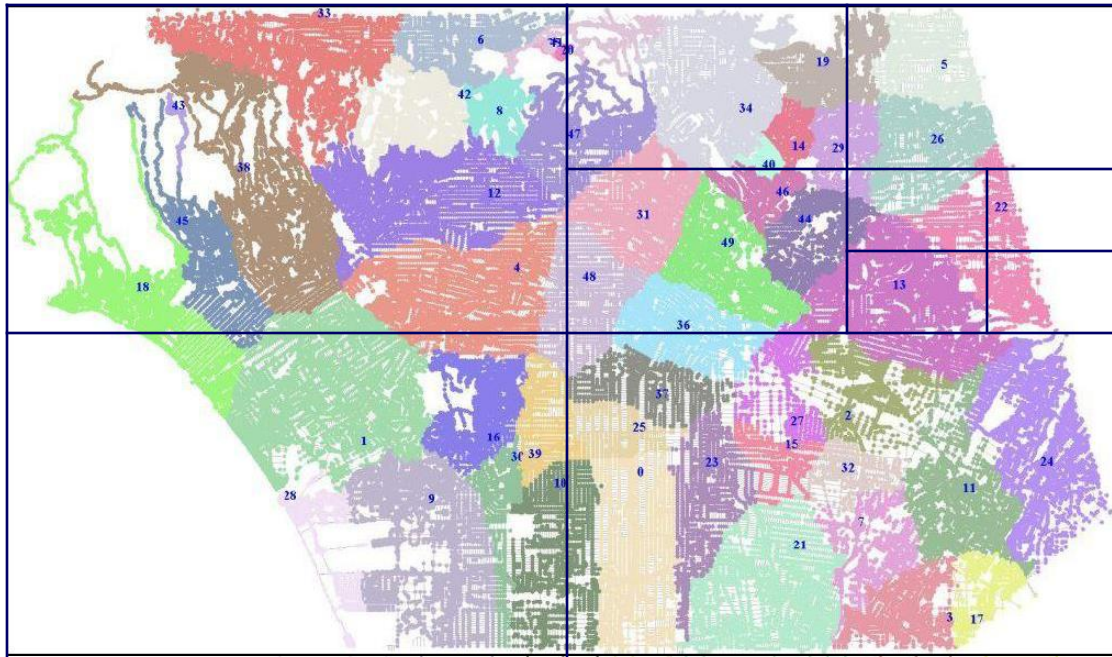
Indexing NVD

- VQ-tree
 - NVPs are
 - **mutually exclusive** : they do not have any overlapping areas
 - **collectively exhaustive**: every node/edges in the network is associated with at least one generator
 - Disjoint decomposition of the NVPs



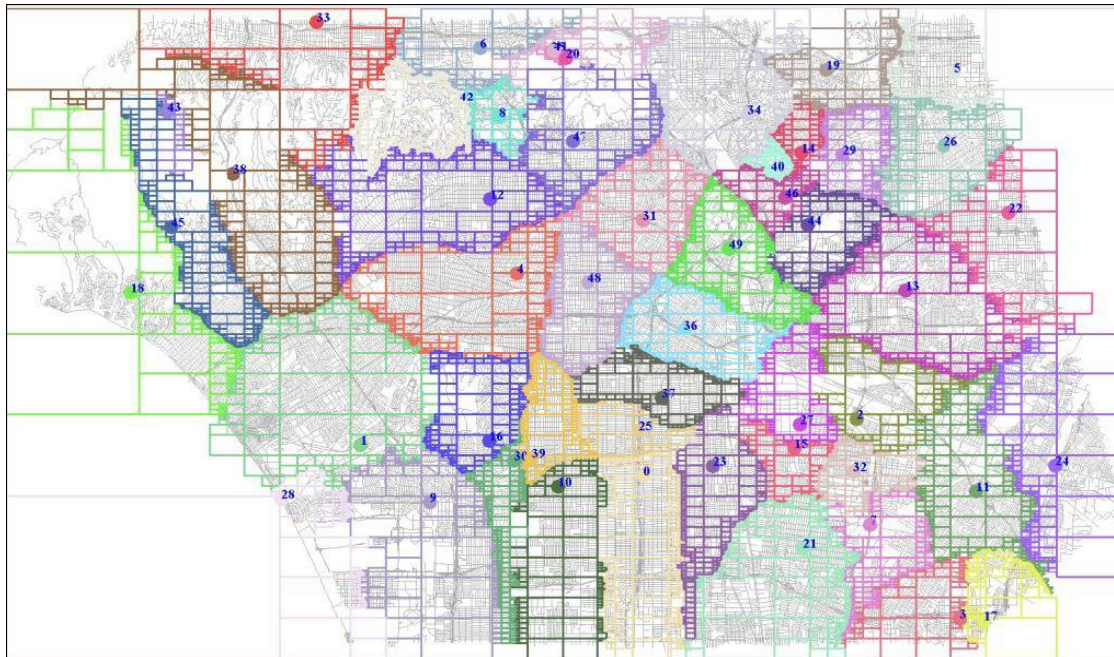
Indexing NVD

- VQ-tree
 - NVPs are
 - **mutually exclusive** : they do not have any overlapping areas
 - **collectively exhaustive**: every node/edges in the network is associated with at least one generator
 - Disjoint decomposition of the NVPs



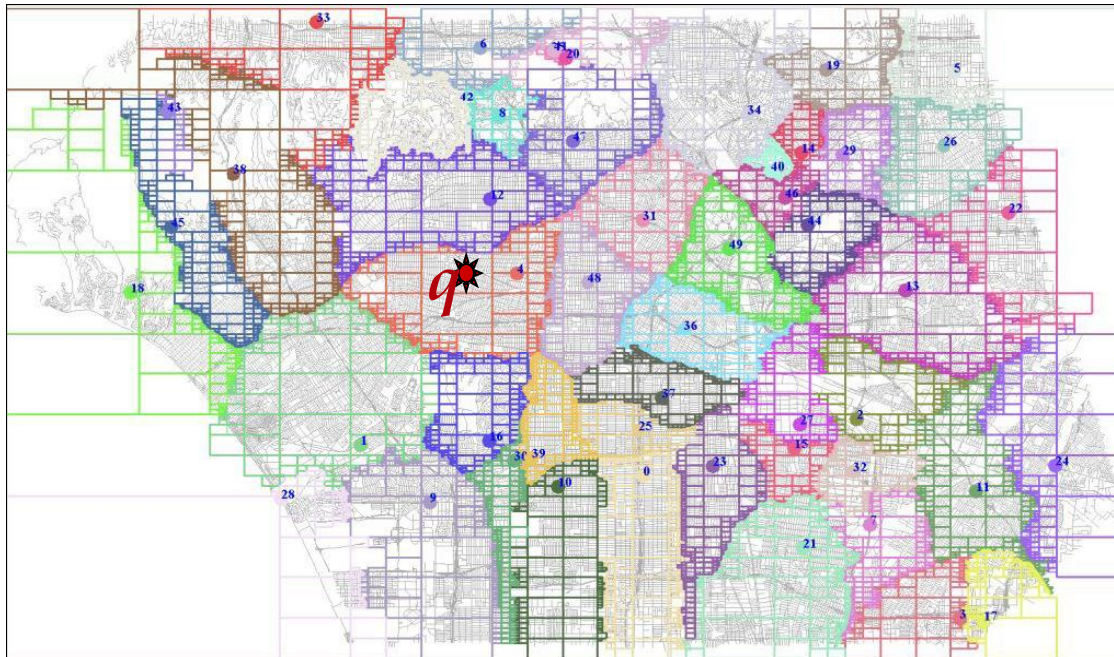
Indexing NVD

- VQ-tree
 - NVPs are
 - **mutually exclusive** : they do not have any overlapping areas
 - **collectively exhaustive**: every node/edges in the network is associated with at least one generator
 - Disjoint decomposition of the NVPs



Indexing NVD

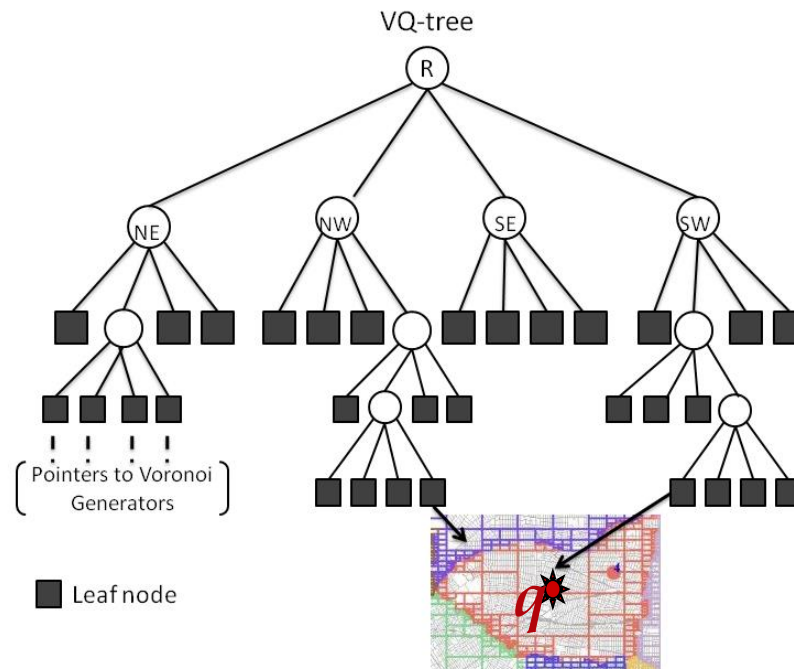
- VQ-tree
 - Each network NVP consists of disjoint quad-tree blocks



Indexing NVD

- VQ-tree

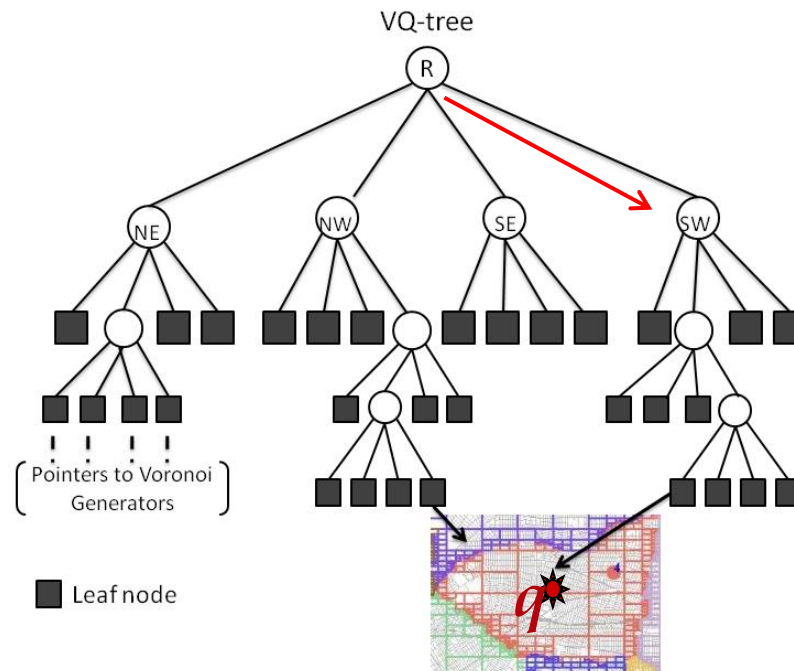
- Each network NVP consists of disjoint quad-tree blocks
- Leaf nodes does not keep any information about the network but store the region information (i.e., coordinates) of the quad-blocks



Indexing NVD

- VQ-tree

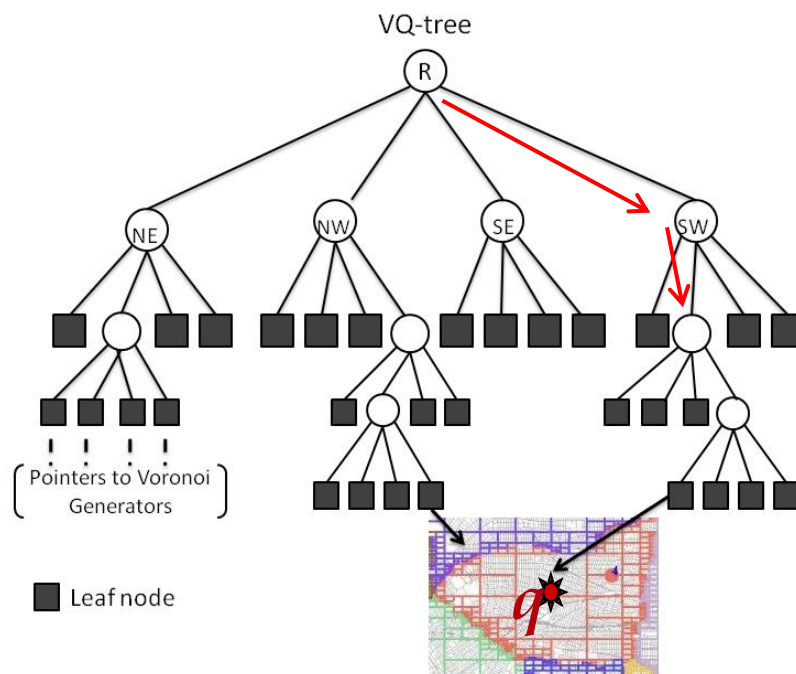
- Each network NVP consists of disjoint quad-tree blocks
- Leaf nodes does not keep any information about the network but store the region information (i.e., coordinates) of the quad-blocks



Indexing NVD

- VQ-tree

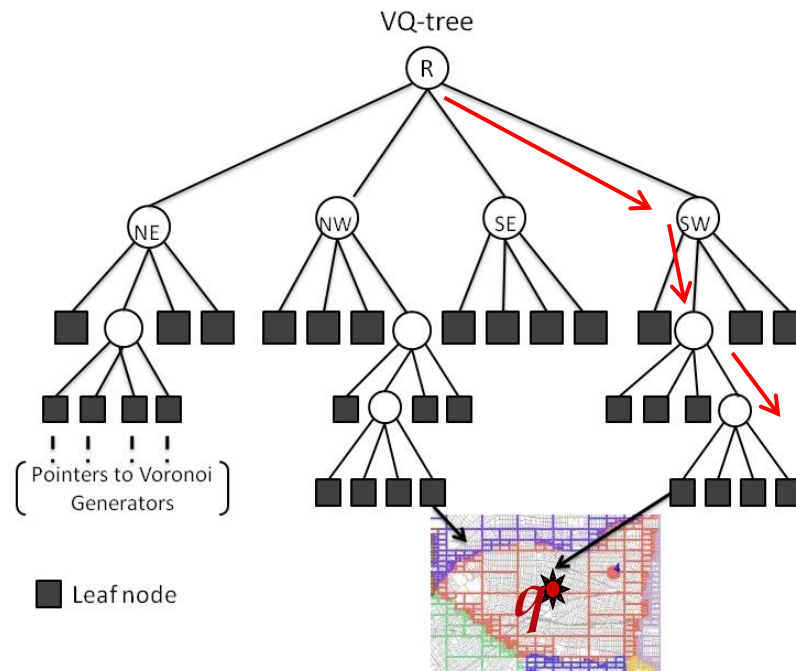
- Each network NVP consists of disjoint quad-tree blocks
- Leaf nodes does not keep any information about the network but store the region information (i.e., coordinates) of the quad-blocks



Indexing NVD

- VQ-tree

- Each network NVP consists of disjoint quad-tree blocks
- Leaf nodes does not keep any information about the network but store the region information (i.e., coordinates) of the quad-blocks





Discussion

- What if the edge weights are changing?
- What if the both query and data objects are moving?
- What if you like to find the nearest hotel and gas station at the same time?



References

- Kolahdouzan, M. R. & Shahabi, C. Voronoi-Based K Nearest Neighbor Search for Spatial Network Databases . VLDB, 2004, 840-851
- A presentation by Ugur Demiryurek in csci587 Fall'2010